

A gradient-enhanced physics-informed neural network (gPINN) scheme for the coupled non-fickian/non-fourierian diffusion-thermoelasticity analysis: A novel gPINN structure

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ABSTRACT

This paper proposes a modified artificial intelligence (AI) approach based on the gradient-enhanced physics-informed neural network (gPINN) with a novel structure for the generalized coupled non-Fickian/non-Fourierian diffusion-thermoelasticity analysis. Previous successful application of gPINN in function approximation and single partial differential equation (PDE) problems gave us enough motivation to develop its application with a novel structure for solving a system of coupled PDEs. The governing equations are formulated for a copper-made strip based upon the Lord-Shulman theory of the generalized coupled thermoelasticity. In order to shed light on the capacity of the gPINN-based approach, three examples with different boundary conditions for a one-dimensional half-space are presented, as well as an example for a two-dimensional half-space. The primary goal is to investigate the transient behavior of field variables (i.e., non-dimensional molar concentration, non-dimensional displacement, and non-dimensional temperature). Variations of dimensionless stress are also examined in depth. Performance comparisons between the gPINN-based method and commonly used approaches support the efficacy of the gPINN with a novel structure. Detailed sensitivity analyses are provided to sufficiently explore the impact of neural network hyperparameters. Moreover, the effect of relaxation time on the behavior of mass concentration is addressed in a single scenario. Overall, the proposed gPINN-based method with a novel structure yields consistently encouraging and satisfactory results, and this approach entails the capability to be applied to more complicated problems. Numerical results underline the importance of the weights assigned to the derivatives of the loss terms to achieve more accurate predictions. Nonparametric statistical tests offer further evidence for exceptional agreement between gPINN solutions with certain weights and outcomes derived by commonly used analytical techniques.

1. Introduction

Nowadays, the applications of methods based on artificial intelligence (AI) in the analysis of some problems in engineering and science have been developed by numerous researchers. Artificial intelligence is a state-of-the-art concept that has been receiving much attention due to its applicability to a wide range of problems. Depending on data availability, problems can be categorized into three types:

- Supervised learning (labeled data is provided);
- Semi-supervised learning (a limited number of labeled data is presented); and
- Unsupervised learning (no labeled data is supplied).

The term supervised learning is generally used in engineering fields to refer to the data-driven tools employed to predict outcomes. Several valuable studies have been presented regarding data-driven approaches (Maghami and Hosseini, 2021; Javadi et al., 2021; Maghami and Hosseini, 2022). One of the widely used and important applications of AI is the utilization of AI-based (unsupervised learning) methods to solve differential equations. In many problems in engineering and science, the governing equations of the problems can be found in the partial differential equations (PDEs) forms. So, the applications of AI-based unsupervised learning methods (especially deep learning (DL) methods) for solving the PDEs have attracted many researchers. Utilizing deep learning to solve PDEs has increased at an unprecedented rate over the last two decades. Initial work in this field focused primarily on

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developing a trial solution (Lagaris et al., 1998). Although some constructive research into solving differential equations has been done (Effati and Pakdaman, 2010; Baymani et al., 2010, 2015), the concept of the trial solution had been neglected for 20 years. The past four years have seen a renewed importance in employing artificial neural networks (ANNs) consisting of the trial solution to solve a broad range of PDEs.

In a major advance in 2020, Samaniego et al. (2020), introduced a novel concept of the deep energy method (DEM). According to (Samaniego et al., 2020), utilization of a potential energy to describe the variations in a given physical body can be a reliable approach as it has been successfully implemented in various computational mechanics problems, including linear elasticity and hyperelasticity (Samaniego et al., 2020), (Nguyen-Thanh et al., 2020), phase-field modelling of fracture, piezoelectricity, and Kirchhoff plate (Samaniego et al., 2020). Parametric deep energy method (P-DEM) (Nguyen-Thanh et al., 2021) is a more recent extension to DEM that addresses the issue of C^1 continuity in strain gradient elasticity. Through conversion of inputs to a reference domain, the common obstacle of vanishing gradients in neural networks was overcome as well.

One of the most effective methods regarding solving PDEs is the physics-informed neural network (PINN), which was initially proposed and employed by Raissi et al. (2019). Since PINN is meshless, it allows efficient computation even in higher dimensions. This method has been widely applied to diverse types of PDEs, namely: stochastic (Yang et al., 2020), (Zhang et al., 2019), fractional (Pang et al., 2019), integro-differential equations (Yuan et al., 2022), (Lu et al., 2021). Moreover, PINN and its variations have covered a surprising number of scientific fields such as physics (Raissi et al., 2019; Yang et al., 2020; Zhang et al., 2019), (Pun et al., 2019; Wang et al., 2021; Rao et al., 2021; Geneva and Zabarar, 2020; Cai et al., 2021), solid mechanics (Cai et al., 2021; Goswami et al., 2020; Haghighat et al., 2021), fluid mechanics (Raissi et al., 2019; Yang et al., 2020; Zhang et al., 2019), (Wang et al., 2021), (Geneva and Zabarar, 2020), (Cai et al., 2021), (Mao et al., 2020; Jin et al., 2021; Kadeethum et al., 2020; Zhu et al., 2019; Yang and Perdikaris, 2019; Wu et al., 2020), thermomechanics (Zobeiry and Humfeld, 2021), (Nguyen et al., 2022), systems biology (Yazdani et al., 2020), geophysics (Kadeethum et al., 2020), (Rasht-Behesht et al., 2022), reliability and safety (Wang et al., 2020), (Zhou et al., 2022), system health monitoring (Liu et al., 2022a), (Shen et al., 2021), control/state estimation (Arnold and King, 2021), designing seismic isolation systems (Habib and Yildirim, 2022), and biomedicine (Sahli Costabal et al., 2020).

In the early stages of development of PINN and its applications in solid mechanics, fundamental advances were made towards enhancing the core capabilities of this DL-based framework. For instance, integration of adaptive collocation method (Anitescu et al., 2019) or deep collocation method (DCM) (Guo et al., 2019) achieved high-resolution results in a wide range of mechanical problems.

A novel architecture incorporating separate neural networks for each output variable, proposed by (Haghighat et al., 2021), is endorsed by its employment on the linear elastostatic and nonlinear elastoplastic problems. In their cutting-edge research, Yu et al. (2022) developed the gradient-enhanced physics-informed neural network (gPINN) to enhance accuracy. In their investigation into considering the gradients of PDE residual with respect to the input variables, Yu et al. (2022) suggest that it can ameliorate the total performance of the neural network. Their claim so far has been applied to simple function approximation and several single forward/inverse PDE problems. A recent paper (Mohammadian et al., 2022) successfully applied gPINN to forward and inverse power system problems, considering scarce data. Several studies have focused on developing supplemental mechanisms with the intention of addressing the shortcomings of the original PINN. Adaptive weight assignment to the loss terms (Wang et al., 2021), (Wang and Zhao, 2021), (Xiang et al., 2022) and meta-learning initialization, such as using the new reptile initialization PINN (NRPINN) (Liu et al., 2022b), are examples of the powerful algorithms incorporated into the

PINN. Using R-functions to reformulate boundary conditions based on distance fields, also called *geometry-aware* PINN (Sukumar and Srivastava, 2022), can accelerate neural network training and improve the accuracy of final predictions. Moreover, a q-stage Runge-Kutta discrete time scheme (Guo et al., 2023) can boost PINN's performance in examining the three-dimensional transient behavior of heat transfer within functionally graded materials (FGMs).

To simulate the behavior of various systems in engineering and science problems, the interactions between various physical fields must be taken into account to have realistic data and results. Considering the interactions between the physical fields in the problems causes the governing equations to be in the systems of the coupled PDEs forms. One of the most important coupled problems in engineering is the coupled thermoelasticity analysis. There are some advanced theories in which the propagation of elastic and thermal waves has been simulated, considering the finite speeds for both elastic and thermal wave propagations. The concepts of the generalized coupled thermoelasticity analysis and the proposed theories in this regard can be found in some very well-known literature, which was presented by Hetnarski and Eslami (2019), Ignaczak and Ostoj-Starzewski (2009), Chandrasekharia (Chandrasekharia, 1998) and other researchers (Shariyat et al., 2010a)– (Zenkour, 2020). In the theories of generalized coupled thermoelasticity analysis, the interactions between the displacement (or stress) field and temperature field are taken into account, under the assumption of finite speeds for both elastic and thermal wave (second sound) propagations. An intriguing problem referred to as coupled diffusion-thermoelasticity analyzes the simultaneous interactions between three field variables: displacement, temperature, and molar concentration. In standard terminology, diffusion is the process of particles moving from areas of high concentration to lower ones (Sherief et al., 2004)– (Sherief and El-Maghraby, 2009). Numerous engineering fields, including geophysics and industry, can benefit from this concept. Initial systematic studies on the theory of thermoelastic diffusion were carried out in 1974 by Nowacki, 1974a, 1974b, 1974c, 1974d. A serious drawback of Fick's law in diffusion is that it fails to take account of the interactions between the given substance and its medium.

A number of authors (Sherief et al., 2004), (Sherief and Saleh, 2005), (Suo and Shen, 2013; Hosseini et al., 2013, 2014a, 2014b, 2015, 2016) challenged this limitation and embarked on the non-Fickian diffusion and thermoelastic analysis. In a study, Sherief et al. (Sherief et al., 2004) derived the governing equations for generalized thermo-diffusion in an elastic body and obtained finite speed propagation for waves. Following this, another research (Sherief and Saleh, 2005) addressed a problem of thermoelasticity in a half-space subjected to a thermal shock loading in the form of the Heaviside unit step function.

Subsequently, some related studies such as incorporation of the magnetic field into two-temperature theory of thermoelasticity (2 TT) (Ezzat and Bary, 2009), Magnetohydrodynamics (MHD) mixed convection flow with the relaxation time and electric permeability of the electromagnetic field (Ezzat et al., 2011), thermal piezo-electric problems with the time-fractional orders (Ezzat and El-Bary, 2012), (Ezzat and El-Bary, 2016) or under photo-excitation transport process (Khamis et al., 2021) have been conducted to cover more practical topics in the area of thermoelasticity.

In order to determine a fully coupled multi-field formulation, the variational principle is adopted. Suo et al. (Suo and Shen, 2012) presented variational principles for coupled temperature-diffusion-mechanics with their set of boundary conditions (BCs) with which numerical solutions are accompanied to highlight the finite speed of wave propagation. An analytical method (Suo and Shen, 2013) applied to a one-dimensional coupled non-Fick diffusion-mechanics problem to find the solution.

In another research, Hosseini et al. (2014a) considered the Green-Naghdi (GN) theory without energy dissipation for a two-dimensional system of coupled non-Fick diffusion thermoelasticity. Hosseini et al. (2016) developed their solution approach following the

above mentioned research. They employed it for a system of generalized coupled non-Fickian diffusion thermoelasticity PDEs to show the dynamic nature of three dependent variables: mass concentration, displacement, and temperature. Several applications of generalized thermoelastic diffusion in the literature are defined in two or three dimensions (Deswal and Choudhary, 2008)–(Youssef et al.). Additionally, some studies analyzed the problem of thermoelastic diffusion on geometries other than the half-space (Sherief and Saleh, 2005), (Elhagary, 2013), (He et al., 2015a, 2015b; Tripathi et al., 2015; El-Sayed, 2016; Youssef and Al-Lehaibi, 2018; Li et al., 2022a, 2022b), such as a strip (Hosseini et al., 2016), a thick plate (Sherief and El-Maghraby, 2009), (Elhagary, 2014), and a hollow cylinder (Elhagary, 2011). Another spectrum of studies draws our attention to problems with variable thermal conductivity and diffusivity within the framework of generalized diffusion thermoelasticity (Li et al., 2017a, 2017b). Researchers in the following papers (Li et al., 2017a, 2017b), dedicate their efforts to investigate the transient response of the concentration, displacement, temperature, stress, and chemical potential under the previously mentioned assumption (i.e., variable thermal conductivity and diffusivity). To solve a wide variety of problems defined within the context of the generalized diffusion thermoelasticity, various analytical and numerical approaches, including Laplace and Fourier transform (Sherief and Saleh, 2005), (Sherief and El-Maghraby, 2009), (Suo and Shen, 2013), (Deswal and Choudhary, 2008)–(Elhagary, 2011), hybrid Laplace transform finite element (He et al., 2015a), finite element method (FEM) (Li et al., 2017a, 2017b), meshless local Petrov-Galerkin (MLPG) (Hosseini et al., 2014a), (Hosseini et al., 2015), (Hosseini et al., 2011a), (Hosseini et al., 2011b), Galerkin Finite Element and Newmark Methods (Hosseini, 2009), a third-order Hermitian finite element (Shariyat et al., 2010b), transfinite element approach (Shariyat, 2012), generalized finite difference (GFD) method (Hosseini, 2021), (Hosseini, 2013), finite difference method (FDM) (as a point-collocation FEM), and (Shariyat et al., 2010a) Talbot technique (Hosseini et al., 2016) have been put forward.

A key problem with much of the literature on employing conventional analytical and numerical approaches is that they predominantly suffer from the *curse of dimensionality*. By adopting more spatial dimensions or by adding extra complexity to the governing equations to be more analogous to real-world applications, these approaches lose their efficiency in terms of the accuracy of the solution and computational time. So, it seems that AI-based solution techniques, such as DL-based approaches, can be employed as effective methods with high accuracy to solve the coupled PDEs.

The main contributions of this research paper can be summarized as follows:

- As one of the novelties in the current research, while a modified structure of separate neural networks for each output variable has already been proposed (Haghighat et al., 2021), (Amini Niaki et al., 2021), this is the first time that this framework has been incorporated into gPINN.
- Based on the performance of the gPINN method, its application can be recommended for the generalized coupled non-Fickian/non-Fourierian diffusion-thermoelasticity analysis. As far as we are aware, this is the first time that gPINN is implemented on the generalized coupled non-Fickian/non-Fourierian diffusion-thermoelasticity problem. Based on the Lord-Shulman theory of coupled thermoelasticity, the governing equations for a strip of copper are derived.
- For a comprehensive assessment of gPINNs, three scenarios for a one-dimensional half-space with different boundary conditions are presented and discussed.
- It is imperative to address gPINN's performance in higher dimensions in terms of accuracy and efficiency. Hence, the behavior of a two-dimensional copper-made half-space based on the theory of generalized thermoelastic diffusion is investigated via gPINN.

- Knowledge of neural networks is merely supplied by the governing equations and known solutions at initial/boundary conditions, and no other data within the domain is introduced.

The aim of the present research is thus to find solutions (forward approach) to the system of coupled PDEs concerning the transient behavior of field variables (i.e., mass concentration, displacement, and temperature). A further important question as to whether gPINN defeats conventional numerical methods is responded. Initially, it seemed that the weights of gradient terms were not influential. However, the more careful analysis revealed that selecting an appropriate value for gradient weights can speed up convergence and result in more accurate solutions. gPINN results match earlier findings (Sherief et al., 2004), (Hosseini et al., 2016). We are of the opinion that the results emphasize the validity of our model. Furthermore, in-depth sensitivity analyses on neural network hyperparameters and the impact of relaxation time on the behavior of field variables are entirely discussed. Friedman's test accompanied by post-hoc Nemenyi test is applied as the nonparametric analysis method to further support the similarity of gPINN results to the analytical ones.

This paper is divided into five sections. In the second section, governing equations of the generalized coupled non-Fickian/non-Fourierian diffusion-thermoelasticity are entirely described. The third section outlines the fundamentals of the proposed methodology. The numerical examples and results are discussed in section four. Through different sets of boundary conditions, three scenarios are defined. Additionally, this section takes a more profound look at gPINN formulations, and comprehensive sensitivity analyses are included to reinforce the reasons behind the selection of neural network hyperparameters. A two-dimensional generalized thermoelastic diffusion problem is discussed in the last subsection and gPINN's accuracy is evaluated via L^2 relative error. The final section deals with the conclusions, research findings, and issues regarding future studies.

2. Mathematical formulations of the problem

The governing equations for the coupled non-Fickian diffusion-thermoelasticity are defined as follows (Sherief et al., 2004), (Sherief and Saleh, 2005), (Hosseini et al., 2016):

$$\sigma_{ij} = 2\mu e_{ij} + \delta_{ij}(\lambda e_{kk} - \beta_1 \theta - \beta_2 c), \quad (1)$$

$$P = -\beta_1 e_{kk} + bc - a\theta \quad (2)$$

In Eqs. (1) and (2), σ_{ij} , e_{ij} , θ , c , P , a and b denote the stress, strain, temperature, mass concentration, chemical potential, measure of thermomodification effect, and measure of diffusive effect, respectively. Moreover, μ and λ are identified as Lamé's constants and $\beta_1 = (3\lambda + 2\mu)\alpha_t$ and $\beta_2 = (3\lambda + 2\mu)\alpha_c$, in which α_t and α_c are coefficients of linear thermal expansion and linear diffusion expansion.

As regards the geometry of the given problem, the functions are all defined in spatio-temporal domain (i.e., x and t). Therefore, displacement components are determined as follows:

$$u_x = u(x, t), u_y = u_z = 0. \quad (3)$$

Below, components of strain tensor can be found:

$$e_{xx} = \frac{\partial u}{\partial x}, e_{yy} = e_{zz} = e_{xy} = e_{yz} = e_{zx} = 0. \quad (4)$$

Considering Eq. (4), stress tensor can be formulated into the following components:

$$\sigma = \sigma_{xx} = (\lambda + 2\mu) \frac{\partial u}{\partial x} - \beta_1 (T - T_0) - B_2 c, \quad (5)$$

$$\sigma_{yy} = \sigma_{zz} = \lambda \frac{\partial u}{\partial x} - \beta_1 (T - T_0) - \beta_2 c, \quad (6)$$

$$\sigma_{xy} = \sigma_{yz} = \sigma_{zx} = 0. \quad (7)$$

Here T_0 is the reference temperature.

An isotropic thermoelastic strip subjected to a thermal shock loading is considered in this section, and the lower boundary is assumed to be traction-free (Sherief et al., 2004), (Sherief and Saleh, 2005), (Hosseini et al., 2016). Accordingly, the coupled PDEs of diffusion-thermoelasticity are formulated as follows:

$$(2\mu + \lambda) \frac{\partial^2 u}{\partial x^2} - \beta_1 \frac{\partial \theta}{\partial x} - \beta_2 \frac{\partial c}{\partial x} = \rho \frac{\partial^2 u}{\partial t^2}, \quad (8)$$

$$k \frac{\partial^2 \theta}{\partial x^2} = \rho c_E \frac{\partial \theta}{\partial t} + \rho c_E \tau_0 \frac{\partial^2 \theta}{\partial t^2} + T_0 \beta_1 \frac{\partial^2 u}{\partial x \partial t} + T_0 \beta_1 \tau_0 \frac{\partial^3 u}{\partial x \partial t^2} + a T_0 \frac{\partial c}{\partial t} + a T_0 \tau_0 \frac{\partial^2 c}{\partial t^2}, \quad (9)$$

$$D \beta_2 \frac{\partial^3 u}{\partial x^3} + D a \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial c}{\partial t} + \tau \frac{\partial^2 c}{\partial t^2} - D b \frac{\partial^2 c}{\partial x^2} = 0, \quad (10)$$

where u , x , ρ , t , k , c_E , τ_0 , τ and D refer to the displacement, position, density, time, thermal conductivity, specific heat, thermal relaxation time, diffusion relaxation time, and thermo-diffusion constant, respectively.

By applying non-dimensional variables, a more convenient form of the coupled equations can be achieved:

$$X = c_0 \omega x, t^* = c_0^2 \omega t, \tau^* = c_0^2 \omega \tau, \tau_0^* = c_0^2 \omega \tau_0, \quad (11)$$

$$\sigma_{ij}^* = \frac{\sigma_{ij}}{\lambda + 2\mu}, C = \frac{\beta_2 c}{\lambda + 2\mu}, P^* = \frac{P}{\beta_2}, U = c_0 \omega u,$$

$$c_0^2 = \frac{\lambda + 2\mu}{\rho}, \omega = \frac{\rho c_E}{k}, \theta^* = \frac{\beta_1 (T - T_0)}{\lambda + 2\mu}.$$

The following is the definition of parameters, including c_0^2 and ω :

$$c_0^2 = \frac{(\lambda + 2\mu)}{\rho}, \omega = \frac{\rho c_E}{k}. \quad (12)$$

Taking advantage of non-dimensional variables and eliminating the asterisk, now Eqs. (8–10) can be reformulated as follows:

$$\frac{\partial^2 U}{\partial X^2} - \frac{\partial \theta}{\partial X} - \frac{\partial C}{\partial X} = \frac{\partial^2 U}{\partial t^2}, \quad (13)$$

$$\frac{\partial^2 \theta}{\partial X^2} = \frac{\partial \theta}{\partial t} + \tau_0 \frac{\partial^2 \theta}{\partial t^2} + b_1 \frac{\partial^2 U}{\partial X \partial t} + b_1 \tau_0 \frac{\partial^3 U}{\partial X \partial t^2} + b_2 \frac{\partial C}{\partial t} + b_2 \tau_0 \frac{\partial^2 C}{\partial t^2}, \quad (14)$$

$$\frac{\partial^3 U}{\partial X^3} + b_3 \frac{\partial^2 \theta}{\partial X^2} + b_4 \left(\frac{\partial C}{\partial t} + \tau \frac{\partial^2 C}{\partial t^2} \right) - b_5 \frac{\partial^2 C}{\partial X^2} = 0, \quad (15)$$

where

$$b_1 = \frac{\beta_1^2 T_0}{\rho c_E (\lambda + 2\mu)}, b_2 = \frac{a \beta_1 T_0}{c_E \rho \beta_2}, b_3 = \frac{a (\lambda + 2\mu)}{\beta_1 \beta_2}, \quad (16)$$

$$b_4 = \frac{k (\lambda + 2\mu)}{D \beta_2^2 \rho c_E}, b_5 = \frac{b (\lambda + 2\mu)}{\beta_2^2}.$$

Moreover, components of stress tensor and the chemical potential are represented in non-dimensional form as what follows:

$$\sigma_{xx} = \frac{\partial U}{\partial X} - \theta - C, \quad (17)$$

$$\sigma_{yy} = \sigma_{zz} = \left(1 - \frac{2}{\beta^2} \right) \frac{\partial U}{\partial X} - \theta - C, \quad (18)$$

$$P = b_5 C - \frac{\partial U}{\partial X} - b_3 \theta. \quad (19)$$

For all the examples presented in this paper, the initial conditions are the same:

$$U(X, t=0) = 0, \frac{\partial U(X, t=0)}{\partial t} = 0, \quad (20)$$

$$C(X, t=0) = 0, \frac{\partial C(X, t=0)}{\partial t} = 0,$$

$$\theta(X, t=0) = 0, \frac{\partial \theta(X, t=0)}{\partial t} = 0.$$

3. gPINN methodology

The general formulation of PDE with the solution of $\mathbf{u}(\mathbf{z})$ can be defined as follows:

$$\mathcal{F}(\mathbf{u}(\mathbf{z}); \vartheta) = f(\mathbf{z}), \mathbf{z} \in \Omega, \quad (21)$$

$$\beta(\mathbf{u}(\mathbf{z})) = h(\mathbf{z}), \mathbf{z} \in \partial\Omega, \quad (22)$$

where $\mathcal{F}$, $\mathbf{u}$, and $\mathbf{z}$ denote the differential operator, solution to the given problem, and vector of input variables (i.e., space and time), respectively. f addresses the forcing term. The given PDE in Eq. (21) has deterministic parameters as ϑ and domain of the problem and boundary conditions are defined on Ω and $\partial\Omega$, respectively. β in Eq. (22) formulates boundary conditions which can be Dirichlet, Neumann, Robin, and other known types. h signifies the value of the boundary condition. As its name implies, initial condition (IC) determines state of the problem at $t = 0$. It is consequently possible to consider ICs as a type of boundary conditions (Lu et al., 2021).

As a first step, neural network structure must be specified. Based upon Eq. (21), it is assumed that $\hat{\mathbf{u}}_{\Phi}(\mathbf{z})$ is the output of neural network and an approximation to $\mathbf{u}(\mathbf{z})$. Φ identifies parameters specific to the neural network (i.e., matrix of weights and vector of biases).

$$\hat{\mathbf{u}}_{\Phi}(\mathbf{z}) \approx \mathbf{u}(\mathbf{z}) \quad (23)$$

Training of the neural network occurs through the optimization of these parameters. Governing equations of the problem accompanied by boundary and initial conditions are all embodied within the neural network. Automatic differentiation (AD), an attractive feature of deep learning, enables the exact and fast computation of all the previously mentioned equations.

One of the fundamental components of every neural network is its loss term. The total loss function of gPINN is constituted from the following terms:

$$\mathcal{L}_{Total} = \mathcal{L}_{PDE} + \mathcal{L}_{grad_PDE(k)_j} + \mathcal{L}_{BC} + \mathcal{L}_{IC} + \mathcal{L}_{Data}, \quad (24)$$

Here in Eq. (24), $\mathcal{L}_{PDE}$ signifies the loss of governing equations calculated on collocation points. $\mathcal{L}_{grad_PDE(k)_j}$ determines the loss term of the gradient of k^{th} PDE in a coupled system with respect to dimension j . $\mathcal{L}_{BC}$ and $\mathcal{L}_{IC}$ compute losses associated with boundary and initial conditions, respectively. In cases where measurement data is available, $\mathcal{L}_{Data}$ is incorporated into the total loss term and identifies the deviations from actual values. $\mathcal{L}_{Total}$ is the summation of the previously mentioned losses.

In the original PINN, merely PDE residuals are considered being zero. When the residuals of a PDE are equal to zero, then it can be justified that the PDE derivatives are also zero (Yu et al., 2022). Providing that the exact solutions are smooth enough, derivatives of PDEs are assumed to exist (Yu et al., 2022).

To offer more information on how to compute loss terms, detailed formulations are presented in the following:

$$\mathcal{L}_{PDE}(\mathbf{x}; \Phi) = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \mathcal{F}(\hat{\mathbf{u}}_{\Phi}(z_i) - f(z_i)) \right\|_2^2, \quad (25)$$

Here $\mathcal{L}_{PDE}$ enforces PDE residuals to be zero on given collocation points (i.e., N_c).

$$\mathcal{L}_{grad_PDE(k)} = w_{grad(k)} \frac{1}{N_D} \sum_{z \in N_D} \left\| \frac{\partial \mathcal{F}}{\partial z_j} \right\|_2^2, \quad (26)$$

The loss term in Eq. (26) imposes derivatives of PDEs to be zero on collocation points denoted by N_D . N_c and N_D can be mutual or separate sets of collocation points. $w_{grad(k)}$ is a loss weight of PDE gradients that sometimes has the capacity to impact convergence rate and solution accuracy. These weights can have different values (Yu et al., 2022). This paper opted for several weights to discover the ideal value.

To compute loss of BCs/ICs, following term summarizes it:

$$\mathcal{L}_\beta(\mathbf{x}; \Phi) = \frac{1}{N_\beta} \sum_{i=1}^{N_\beta} \left\| \beta(\hat{u}_\Phi(z_i) - h(z_i)) \right\|_2^2. \quad (27)$$

Here N_β denotes BC/IC points. It is often practical to add $\mathcal{L}_{data}$ provided that measurement data (experimental or synthetic) is available:

$$\mathcal{L}_{data}(\mathbf{x}; \Phi) = \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} \left\| \hat{u}_\Phi(z_i) - u_i^* \right\|_2^2, \quad (28)$$

where N_{data} is the number of observations collected via the experiments or synthetic simulations.

A crucial requirement to achieve the unique solution of a PDE is to define boundary conditions. Early studies of PINN enforce BCs and ICs as soft constraints (Raissi et al., 2019), (Yazdani et al., 2020), (Raissi et al., 2020). The main downside of this procedure is that BCs and ICs are not generally fully satisfied, which sometimes leads to higher errors in predicting the solution. The idea of hard BC was first proposed by Lagaris et al. (1998) to enforce Dirichlet BCs as hard constraints. In recent years, the imposition of other types of BCs has been investigated (Lagari et al., 2020). Through the use of simple polynomials, desired BCs/ICs are completely satisfied. This paper analyzes three scenarios with different sets of BCs, and trial solutions are constructed for all these

Here, it is assumed that p_1 and θ_1 are both equal to one (Sherief and Saleh, 2005).

In order to study the behavior of the system of coupled PDEs, the copper material has been selected, and the associated parameters are described below in SI units. This enabled us to compare the gPINN results with previous studies in the literature (Sherief and Saleh, 2005), (Hosseini et al., 2016).

$$\lambda = 7.76 \times 10^{10} \left(\frac{N}{m^2} \right) \mu = 3.86 \times 10^{10} \left(\frac{N}{m^2} \right) \rho = 8954 \left(\frac{kg}{m^3} \right) D = 0.85 \times 10^{-8} \left(\frac{kg \cdot s}{m^3} \right)$$

$$b = 0.9 \times 10^6 \left(\frac{m^5}{kg \cdot s^2} \right) \alpha_c = 1.98 \times 10^{-4} \left(\frac{m^3}{kg} \right) \tau_0 = 0.02 \alpha_r = 1.78 \times 10^{-5} \left(\frac{1}{K} \right)$$

$$k = 386 \left(\frac{W}{m \cdot K} \right) a = 1.2 \times 10^4 \left(\frac{m^2}{s^2 \cdot K} \right) \tau = 0.2 c_E = 383.1 \left(\frac{J}{kg \cdot K} \right)$$

$$c_1 = 1 \quad T_0 = 300 (^{\circ}K) \quad \theta_1 = 1$$

According to the aforementioned values of the parameters, the coefficients in Eqs. (13–15) are computed as follows:

$$b_1 = 0.0168 \quad b_2 = 0.0921 \quad b_3 = 5.4846 \quad b_4 = 0.5439 \quad b_5 = 36.9794$$

The geometry of the example is defined on a strip of copper subjected to the thermal and mass concentration shock loading on the inner boundary (see Fig. 1). Considering that the three output variables have transient behavior, the best approach would be to apply gPINN with separate networks.

To be more precise, an approximation to every output variable (i.e., $\hat{C}(x, t; \Phi)$, $\hat{U}(x, t; \Phi)$, and $\hat{\theta}(x, t; \Phi)$) has its own neural network containing a specific number of hidden layers and neurons. The total loss function (i.e., $\mathcal{L}_{Total}$) is composed of the following terms:

$$\mathcal{L}_{Total} = \mathcal{L}_{PDE(1)} + \mathcal{L}_{PDE(2)} + \mathcal{L}_{PDE(3)} + \mathcal{L}_{grad_PDE(1)_{x,t}} + \mathcal{L}_{grad_PDE(2)_{x,t}} + \mathcal{L}_{grad_PDE(3)_{x,t}} + \mathcal{L}_{C_0} + \mathcal{L}_{U_0} + \mathcal{L}_{\theta_0} + \mathcal{L}_{C'_0} + \mathcal{L}_{U'_0} + \mathcal{L}_{\theta'_0} + \mathcal{L}_{C_{BC(1)}} + \mathcal{L}_{C_{BC(2)}} + \mathcal{L}_{U_{BC(1)}} + \mathcal{L}_{U_{BC(2)}} + \mathcal{L}_{\theta_{BC(1)}} + \mathcal{L}_{\theta_{BC(2)}}, \quad (30)$$

three examples.

Furthermore, neural network performance is dependent upon the choice of hyperparameters, including the number of layers (also known as depth of neural network), number of hidden neurons (width of neural network), learning rate, number of iterations, batch size, the structure of neural networks, and other related components. To adjust the efficiency of gPINN to its satisfactory level, several sets of sensitivity analysis on hyperparameters are conducted.

4. Numerical examples and results

4.1. First example

The following BCs are taken into consideration as the first example. This allows the comparison and verification of gPINN results with conventional approaches (Sherief and Saleh, 2005).

$$\sigma(X=0, t) = 0, \sigma(X=1.0, t) = 0, \quad (29)$$

$$P(X=0, t) = p_1 H(t), P(X=1.0, t) = 0,$$

$$\theta(X=0, t) = \theta_1 H(t), \theta(X=1.0, t) = 0.$$

where $\mathcal{L}_{PDE(1)}$, $\mathcal{L}_{PDE(2)}$, and $\mathcal{L}_{PDE(3)}$ stand for the loss terms associated with the residuals of each PDE in the coupled system (see Eqs. (13–15)). $\mathcal{L}_{grad_PDE(1)_{x,t}}$, $\mathcal{L}_{grad_PDE(2)_{x,t}}$, and $\mathcal{L}_{grad_PDE(3)_{x,t}}$ denote the losses for

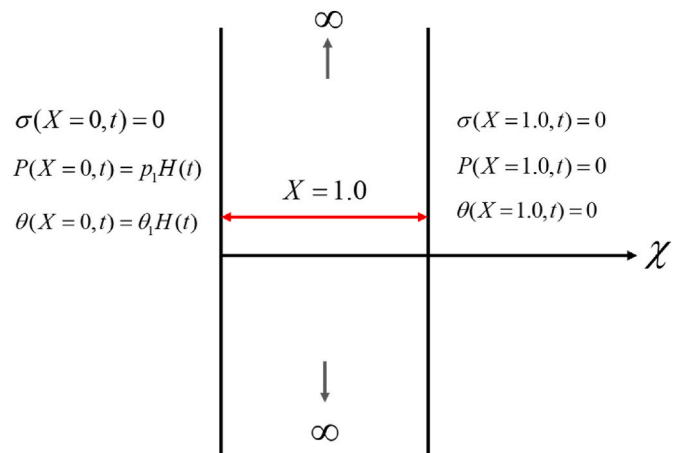


Fig. 1. Schematic of the geometry of the problem as a copper-made strip.

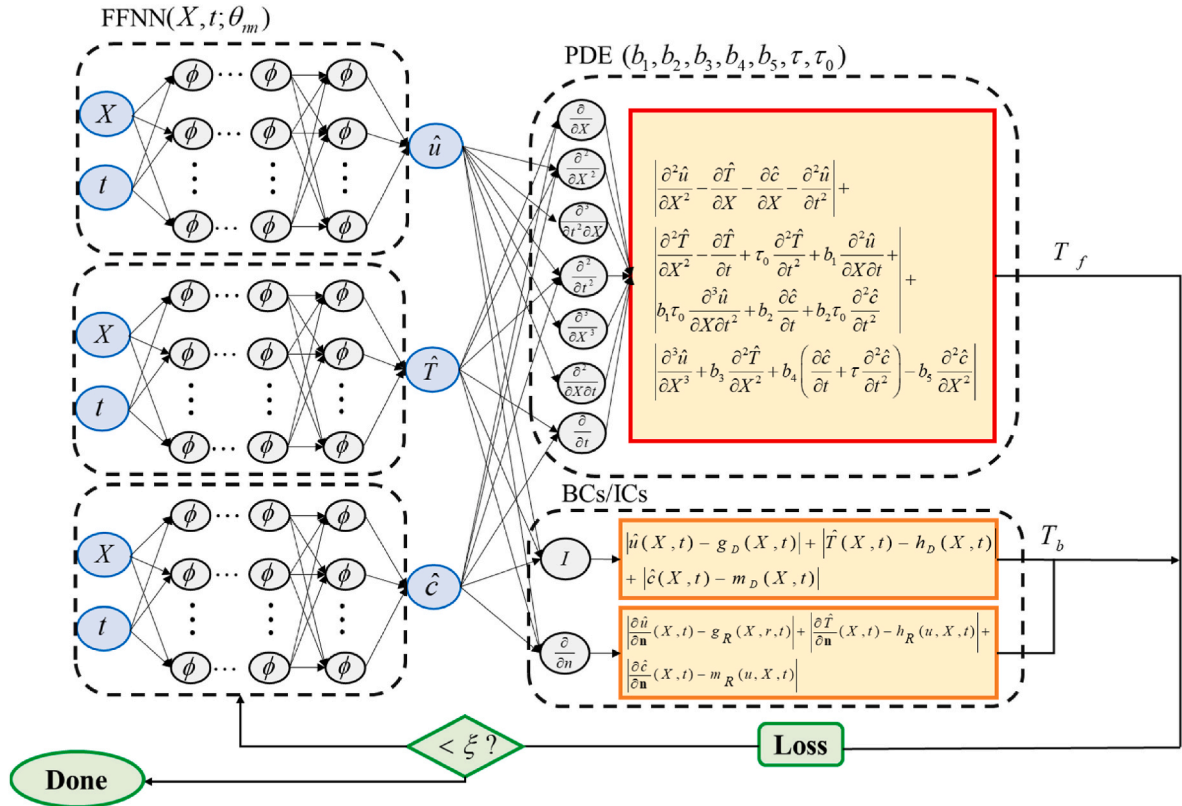


Fig. 2. General structure of the original PINN for the generalized coupled non-Fickian/non-Fourierian diffusion-thermoelasticity problem.

gradients of PDEs with respect to x and t . As depicted in Fig. 2, the original PINN does not take into account these six terms of PDE gradients. $\mathcal{L}_{C_0}$ and $\mathcal{L}_{C_0}$ identify the losses of simple and Neumann initial conditions for the mass concentration, respectively. The same procedure applies to the initial conditions of other field variables (i.e., $\mathcal{L}_{U_0}$, $\mathcal{L}_{U_0}$, $\mathcal{L}_{\theta_0}$, and $\mathcal{L}_{\theta_0}$). $\mathcal{L}_{C_{BC(1)}}$ defines the loss of boundary condition for mass concentration at the inner boundary and similarly, $\mathcal{L}_{C_{BC(2)}}$ signifies the loss of boundary condition at the outer boundary. Boundary conditions for two other field variables (i.e., U and θ) are defined the same as C .

For the sake of clarity, details of the loss terms associated with the first scenario are provided:

$$\mathcal{L}_{PDE(1)} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \frac{\partial^2 U}{\partial X^2} - \frac{\partial \theta}{\partial X} - \frac{\partial C}{\partial X} - \frac{\partial^2 U}{\partial t^2} \right\|_2^2, \quad (31)$$

$$\mathcal{L}_{PDE(2)} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \frac{\partial^2 \theta}{\partial X^2} - \frac{\partial \theta}{\partial t} - \tau_0 \frac{\partial^2 \theta}{\partial t^2} - b_1 \frac{\partial^2 U}{\partial X \partial t} - b_1 \tau_0 \frac{\partial^3 U}{\partial X \partial t^2} - b_2 \frac{\partial C}{\partial t} - b_2 \tau_0 \frac{\partial^2 C}{\partial t^2} \right\|_2^2, \quad (32)$$

$$\mathcal{L}_{PDE(3)} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \frac{\partial^3 U}{\partial X^3} + b_3 \frac{\partial^2 \theta}{\partial X^2} + b_4 \left(\frac{\partial C}{\partial t} + \tau \frac{\partial^2 C}{\partial t^2} \right) - b_5 \frac{\partial^2 C}{\partial X^2} \right\|_2^2, \quad (33)$$

$$\mathcal{L}_{grad_PDE(1)_x} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^3 U}{\partial X^3} - \frac{\partial^2 \theta}{\partial X^2} - \frac{\partial^2 C}{\partial X^2} - \frac{\partial^3 U}{\partial X \partial t^2} \right\|_2^2, \quad (34)$$

$$\mathcal{L}_{grad_PDE(1)_t} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^3 U}{\partial X^2 \partial t} - \frac{\partial^2 \theta}{\partial X \partial t} - \frac{\partial^2 C}{\partial X \partial t} - \frac{\partial^3 U}{\partial t^3} \right\|_2^2, \quad (35)$$

$$\mathcal{L}_{grad_PDE(2)_x} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^3 \theta}{\partial X^3} - \frac{\partial^2 \theta}{\partial X \partial t} - \tau_0 \frac{\partial^3 \theta}{\partial X \partial t^2} - b_1 \frac{\partial^3 U}{\partial X^2 \partial t} - b_1 \tau_0 \frac{\partial^4 U}{\partial X^2 \partial t^2} - b_2 \frac{\partial^2 C}{\partial X \partial t} - b_2 \tau_0 \frac{\partial^3 C}{\partial X \partial t^2} \right\|_2^2, \quad (36)$$

$$\mathcal{L}_{grad_PDE(2)_t} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^3 \theta}{\partial X^2 \partial t} - \frac{\partial^2 \theta}{\partial t^2} - \tau_0 \frac{\partial^3 \theta}{\partial t^3} - b_1 \frac{\partial^3 U}{\partial X \partial t^2} - b_1 \tau_0 \frac{\partial^4 U}{\partial X \partial t^3} - b_2 \frac{\partial^2 C}{\partial t^2} - b_2 \tau_0 \frac{\partial^3 C}{\partial t^3} \right\|_2^2, \quad (37)$$

$$\mathcal{L}_{grad_PDE(3)_x} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^4 U}{\partial X^4} + b_3 \frac{\partial^3 \theta}{\partial X^3} + b_4 \left(\frac{\partial^2 C}{\partial X \partial t} + \tau \frac{\partial^3 C}{\partial X \partial t^2} \right) - b_5 \frac{\partial^3 C}{\partial X^3} \right\|_2^2, \quad (38)$$

$$\mathcal{L}_{grad_PDE(3)_t} = \frac{1}{N_D} \sum_{i=1}^{N_D} \left\| \frac{\partial^4 U}{\partial X^3 \partial t} + b_3 \frac{\partial^3 \theta}{\partial X^2 \partial t} + b_4 \left(\frac{\partial^2 C}{\partial t^2} + \tau \frac{\partial^3 C}{\partial t^3} \right) - b_5 \frac{\partial^3 C}{\partial X^2 \partial t} \right\|_2^2, \quad (39)$$

$$\mathcal{L}_{C_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \hat{C}(X, t=0) \right\|_2^2, \quad (40)$$

$$\mathcal{L}_{U_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \hat{U}(X, t=0) \right\|_2^2, \quad (41)$$

$$\mathcal{L}_{\theta_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \hat{\theta}(X, t=0) \right\|_2^2, \quad (42)$$

$$\mathcal{L}_{C_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \frac{\partial \widehat{C}}{\partial t}(X, t=0) \right\|_2^2, \quad (43)$$

$$\mathcal{L}_{U_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \frac{\partial \widehat{U}}{\partial t}(X, t=0) \right\|_2^2, \quad (44)$$

$$\mathcal{L}_{\theta_0} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left\| \frac{\partial \widehat{\theta}}{\partial t}(X, t=0) \right\|_2^2, \quad (45)$$

$$\mathcal{L}_{C_{BC(1)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| b_5 \widehat{C}(X=0, t) - \frac{\partial \widehat{U}}{\partial X}(X=0, t) - b_3 \widehat{\theta}(X=0, t) - p_1 H(t) \right\|_2^2, \quad (46)$$

$$\mathcal{L}_{C_{BC(2)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| b_5 \widehat{C}(X=1.0, t) - \frac{\partial \widehat{U}}{\partial X}(X=1.0, t) - b_3 \widehat{\theta}(X=1.0, t) \right\|_2^2, \quad (47)$$

$$\mathcal{L}_{U_{BC(1)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| \frac{\partial \widehat{U}}{\partial X}(X=0, t) - \widehat{\theta}(X=0, t) - \widehat{C}(X=0, t) \right\|_2^2, \quad (48)$$

$$\mathcal{L}_{U_{BC(2)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| \frac{\partial \widehat{U}}{\partial X}(X=1.0, t) - \widehat{\theta}(X=1.0, t) - \widehat{C}(X=1.0, t) \right\|_2^2, \quad (49)$$

$$\mathcal{L}_{\theta_{BC(1)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| \widehat{\theta}(X=0, t) - \theta_1 H(t) \right\|_2^2, \quad (50)$$

$$\mathcal{L}_{\theta_{BC(2)}} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left\| \widehat{\theta}(X=1.0, t) \right\|_2^2. \quad (51)$$

In order to obtain the optimum values of the weight matrices and bias vectors, minimization must be undertaken with respect to three sets of parameters (i.e., Φ_C , Φ_U , and Φ_θ). Consequently, it is evident that gPINN is a multi-objective optimization problem. Total number of points required to establish the neural network is $N_{total} = N_{C/D} + N_{IC} + N_{BC}$. Here it is assumed that $N_c = N_D$, and it is therefore counted once. A comparative analysis between an individual neural network and multiple neural networks is presented to address systematically the probable capacities of the latter structure.

Multiple independent runs with a range of assumptions are needed to derive the architecture of gPINN and hyperparameter setting. Table 1 provides details regarding the effect of the number of layers, hidden neurons, learning rate, number of epochs, and separate or unified neural networks.

Various combinations of the hyperparameters are listed alphabetically from A to L, and the corresponding loss is computed in the last column. Table 1 seems to demonstrate that, by adding layers and hidden

neurons, it is required to consider more epochs to reach lower losses. Sets of hyperparameters, including the number of layers = {2, 3, 4, 5}, number of neurons = {20, 30, 40, 50}, learning rate = {1e-4, 1e-3}, epochs = {30,000, 40,000, 50,000}, and single/separate architecture are represented in columns 2 to 6, respectively.

From Table 1, it is clear that scenario L put forward the optimal combination of hyperparameters. Therefore, the results of this example are computed by three separate neural networks, each with five hidden layers and 60 neurons. Also, the learning rate and the number of epochs are equal to 1e-3 and 50,000, respectively. To provide further insight into determining the desired values of hyperparameters, the corresponding box plots are shown in Figs. 3–6.

By imposing boundary conditions as hard constraints, the trial solutions can be described as follows:

$$\widehat{C}(x, t; \Phi) = \|t\|^2 N_c(x, t, \Phi), \quad (52)$$

$$\widehat{U}(x, t; \Phi) = \|t\|^2 N_u(x, t, \Phi), \quad (53)$$

$$\widehat{\theta}(x, t; \Phi) = \|t\|^2 \|1.0 - X\| \|N_T(x, t, \Phi) + \|1.0 - X\| H(t). \quad (54)$$

Here, initial conditions of all three outputs and boundary conditions of the temperature are completely enforced as hard constraints. Since there is a separate neural network for every field variable, each trial solution is composed of its specific neural network.

The propagation of all three field variables, namely mass concentration, displacement, and temperature, is computed using gPINN. The corresponding results can be seen in Figs. 7–9. The results of gPINN corroborate with previous outcomes (Sherief and Saleh, 2005) and (Hosseini et al., 2016) obtained by the Laplace transform and Talbot technique. Moreover, boundary conditions fit well and confirm the propriety of the proposed methodology.

Figs. 7–9 show that the propagation of wave fronts for these three field variables (i.e., concentration, displacement, and temperature) occurs at finite speed. Fig. 8 highlights that the displacement is gradually decreasing and reaching zero as the diffusion distance grows.

Variations of non-dimensional stress at several points in time are demonstrated in Fig. 10. These values fit well with (Sherief and Saleh, 2005) and (Hosseini et al., 2016) and lend support to both the propagation of wave front with finite speed and the predictive capabilities of gPINN.

To widen our knowledge, a set of analyses is conducted to evaluate the impact of relaxation time on wave propagation of non-dimensional mass concentration. Results shown in Fig. 11 reinforce the earlier results (Hosseini et al., 2016) and prove that lowering the relaxation time will decrease the speed of wave propagation in dimensionless mass concentration.

In an attempt to determine whether weights of PDE gradients influence the prediction capabilities of gPINN, this approach is implemented for several weight values, and the results are compared. Hyperparameters' setup for the gPINN is practically the same for all

Table 1
Systematic sensitivity analysis of influential hyperparameters of gPINN in example 1.

Scenario	#Layers	#Neurons	Learning rate	#Epochs	gPINN architecture	Loss
A	2	20	1e-3	30,000	Single	0.0537286
B	2	20	1e-3	30,000	Separate	0.013207841
C	3	20	1e-3	30,000	Single	0.0487
D	3	20	1e-3	30,000	Separate	0.013745767
E	4	30	1e-3	30,000	Single	0.054449579
F	4	30	1e-3	30,000	Separate	0.007996223
G	4	50	1e-4	40,000	Single	0.009248343
H	4	50	1e-4	40,000	Separate	0.002762951
I	5	60	1e-3	40,000	Single	0.005660706
J	5	60	1e-3	40,000	Separate	0.00033918
K	5	60	1e-3	50,000	Single	1.25E-03
L	5	60	1e-3	50,000	Separate	0.000157262

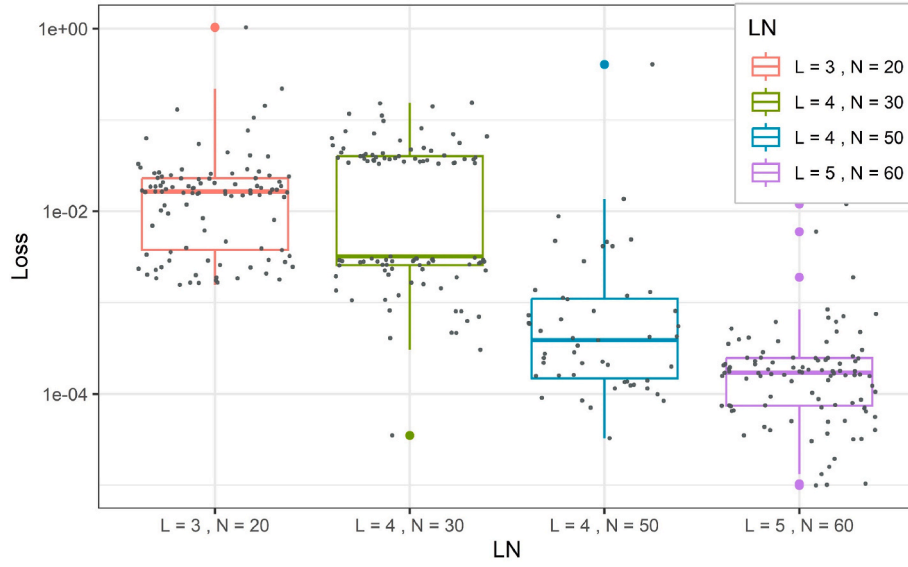


Fig. 3. Boxplot of sensitivity analysis illustrating the effect of number of layers and neurons on the total loss function.

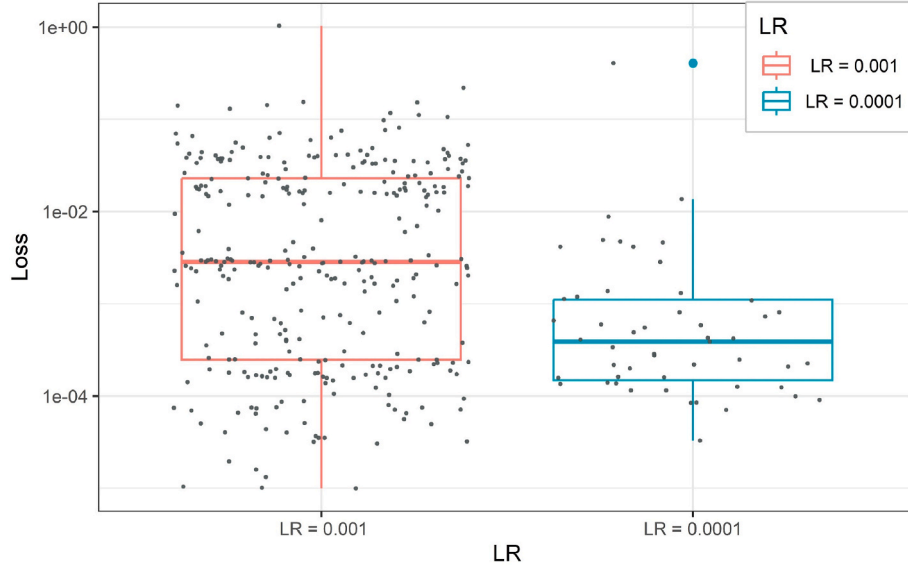


Fig. 4. Boxplot of sensitivity analysis pertaining to the impact of learning rate on the total loss function.

implementations. Six loss terms associated with gradients of PDEs possess weights under this assumption that $w_{grad(1)_{x,t}} = w_{grad(2)_{x,t}} = w_{grad(3)_{x,t}}$. Computational results of the dimensionless concentration, displacement, and temperature are depicted in Figs. 12–14.

As Figs. 12–14 display, gPINN is applied with four different weights. We are surprised to find that there is a substantial disagreement between gPINN results regarding gradient weights. This implies that weights of gradients have an appreciable impact on the accuracy of final results, and $w_{grad(k)_{x,t}} = 0.001$ can lead to reliable solutions.

4.1.1. Statistical analysis

In the previous section, it has been revealed that gPINN with $w_{grad(k)_{x,t}} = 0.001$ achieves satisfactory outcomes. To systematically justify the selection of gPINN structure and its assigned weights (i.e., $w_{grad(k)_{x,t}} = 0.001$), a nonparametric statistical analysis called Friedman's test is conducted.

Nonparametric tests usually refer to statistical analysis methods that do not require certain assumptions about the distribution of the data. In

Friedman's two-way analysis, the null hypothesis states that the response variable follows the same distribution (Pereira et al., 2015). If the null hypothesis is rejected (i.e., p-value < 0.05), it can be deduced that there are significant differences among treatments (Pereira et al., 2015).

- H_0 : the response variable has the same distribution across different groups.
- H_1 : there is a significant difference among the distribution of the groups.

As the null hypothesis is rejected, implementing a post-hoc test would be needed to detect exact treatments with different distributions. The Nemenyi test (also referred to as the Wilcoxon-Nemenyi-McDonald-Thompson test) is adopted. Here, the distributions of the three groups of results are compared: analytical method (Sherief and Saleh, 2005), (Hosseini et al., 2016), gPINN with $w_{grad(k)_{x,t}} = 0.001$, and gPINN with $w_{grad(k)_{x,t}} = 0.1$. Since the gPINN solutions derived from other weights (i.e., $w_{grad(k)_{x,t}} = 0.01$ and $w_{grad(k)_{x,t}} = 1.0$) have approximately similar distributions to the ones obtained via the gPINN with $w_{grad(k)_{x,t}} = 0.1$, they are not included in the Friedman's test. Figs. 15–17 summarize the

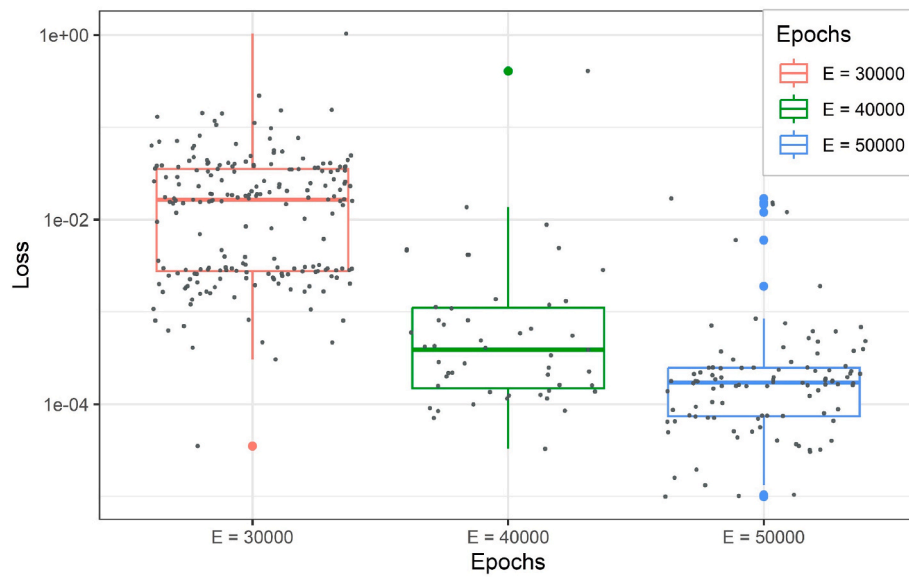


Fig. 5. Boxplot of sensitivity analysis about the number of epochs and its effect on the total loss function.

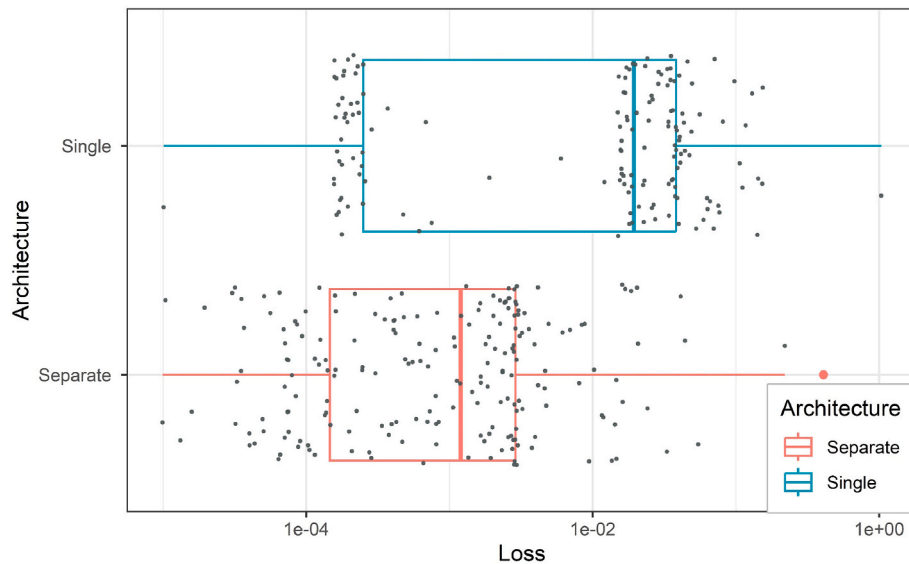


Fig. 6. Boxplot of sensitivity analysis investigating the significance of separate neural networks on reduction of total loss function.

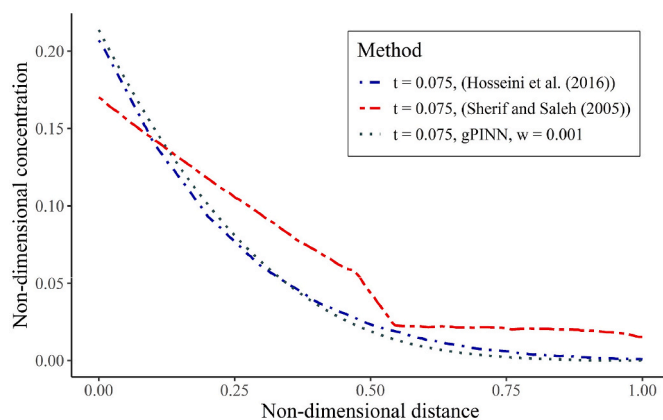


Fig. 7. Comparison of non-dimensional mass concentration obtained by gPINN with those reported in (Sherief and Saleh, 2005) and (Hosseini et al., 2016).

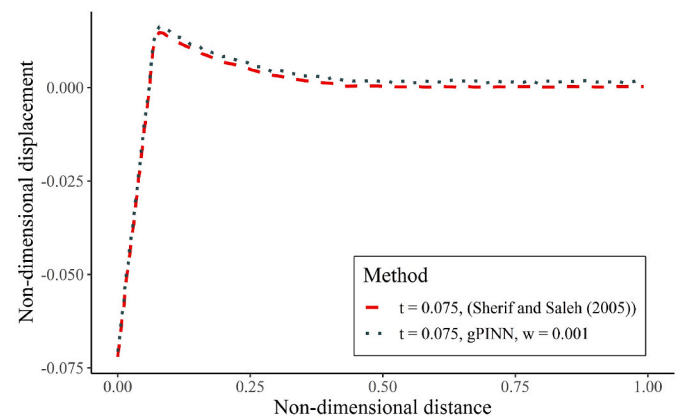


Fig. 8. Comparison of non-dimensional displacement obtained by gPINN with those reported in (Sherief and Saleh, 2005).

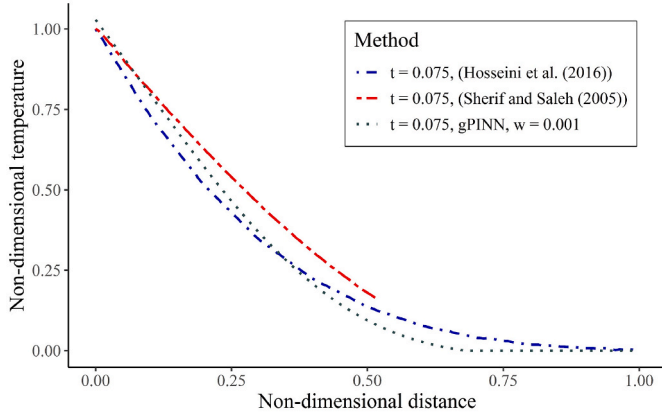


Fig. 9. Comparison of non-dimensional temperature obtained by gPINN with those reported in (Sherief and Saleh, 2005) and (Hosseini et al., 2016).

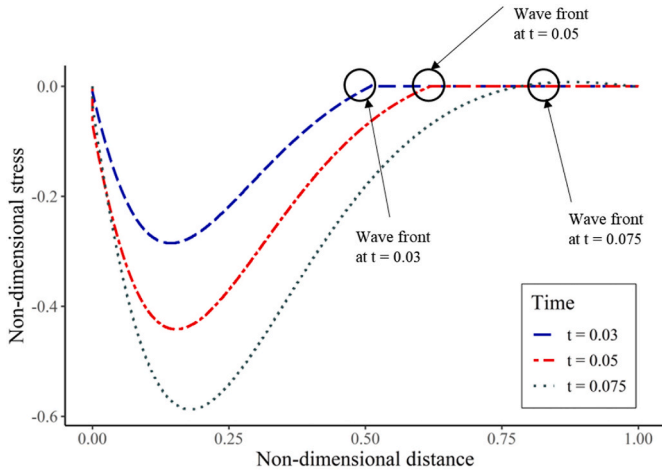


Fig. 10. Variations of non-dimensional stress along X distance.

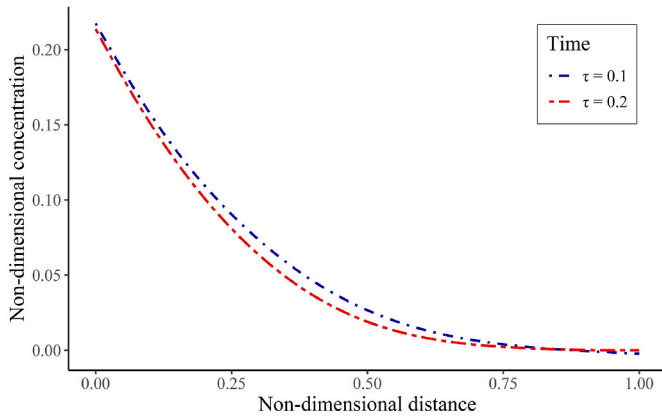


Fig. 11. The impact of relaxation time on velocity of wave propagation in non-dimensional mass concentration.

implementation of Friedman's test accompanied by post-hoc Nemenyi test for concentration, displacement, and temperature, respectively.

As shown in Fig. 15, three groups are compared, namely: A, B, and C. These groups denote the concentration results computed by the analytical method (Hosseini et al., 2016), gPINN with $w_{grad(k)_{x,t}} = 0.001$, and gPINN with $w_{grad(k)_{x,t}} = 0.1$, respectively. Boxplots of two-by-two comparisons on the right-hand side of Fig. 15 prove that there is a

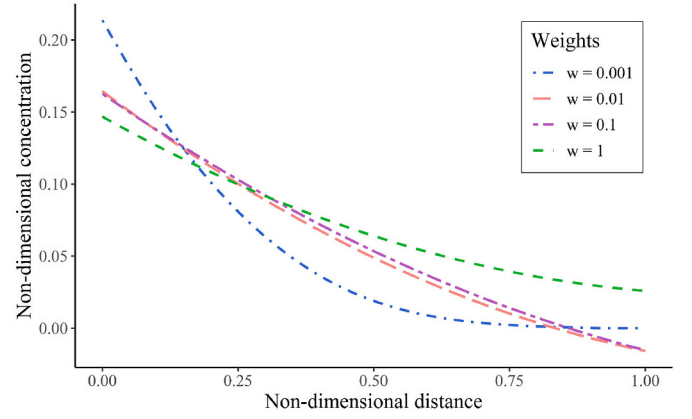


Fig. 12. Comparison of the variations of dimensionless concentration obtained by gPINN with different weights at $t = 0.075$.

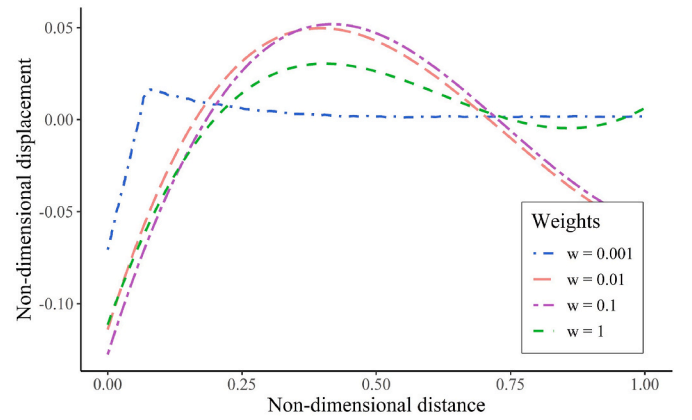


Fig. 13. Comparison of the variations of dimensionless displacement obtained by gPINN with different weights at $t = 0.075$.

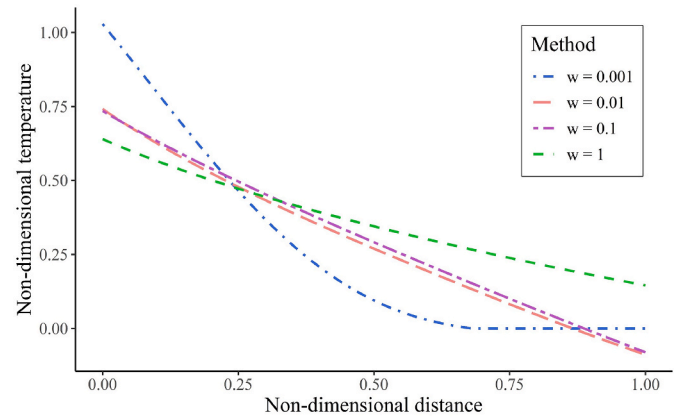


Fig. 14. Comparison of the variations of dimensionless temperature obtained by gPINN with different weights at $t = 0.075$.

significant difference between the distributions of A and C (i.e., the analytical solutions (Hosseini et al., 2016) and gPINN with $w_{grad(k)_{x,t}} = 0.1$). Moreover, the p-values are listed and confirm that gPINN with $w_{grad(k)_{x,t}} = 0.001$ has the same distribution as the analytical method (Hosseini et al., 2016). The parallel coordinates plot on the left-hand side of Fig. 15 provides additional support for the similarity of trends in the concentration results of A and B.

Fig. 16 demonstrates the differences among the displacement

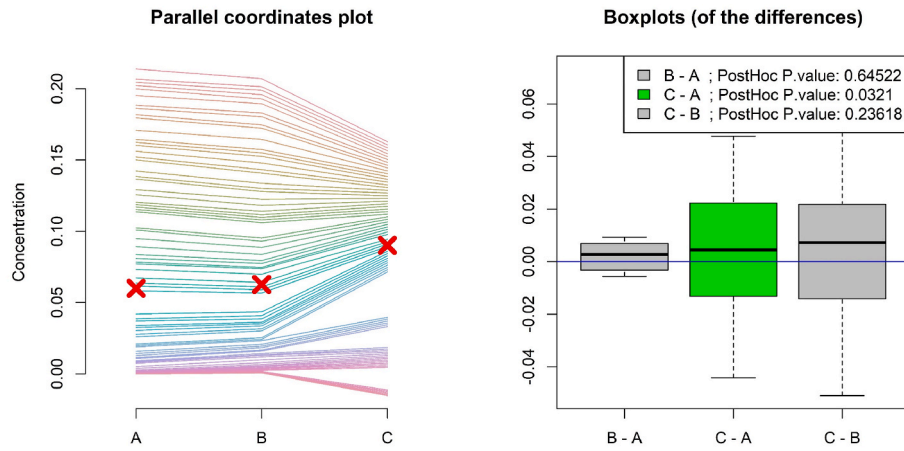


Fig. 15. Results of Friedman's test followed by post-hoc Nemenyi test comparing the concentration solutions obtained via the analytical method (Hosseini et al., 2016) (A), gPINN with $w_{grad(k)_{x,t}} = 0.001$ (B), and gPINN with $w_{grad(k)_{x,t}} = 0.1$ (C).

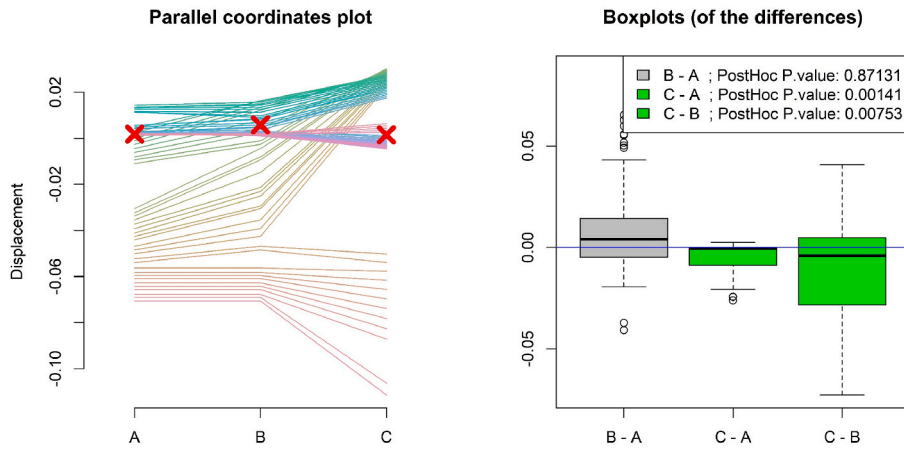


Fig. 16. Results of Friedman's test followed by post-hoc Nemenyi test comparing the displacement solutions calculated via the analytical method (A) (Sherief and Saleh, 2005), gPINN with $w_{grad(k)_{x,t}} = 0.001$ (B), and gPINN with $w_{grad(k)_{x,t}} = 0.1$ (C).

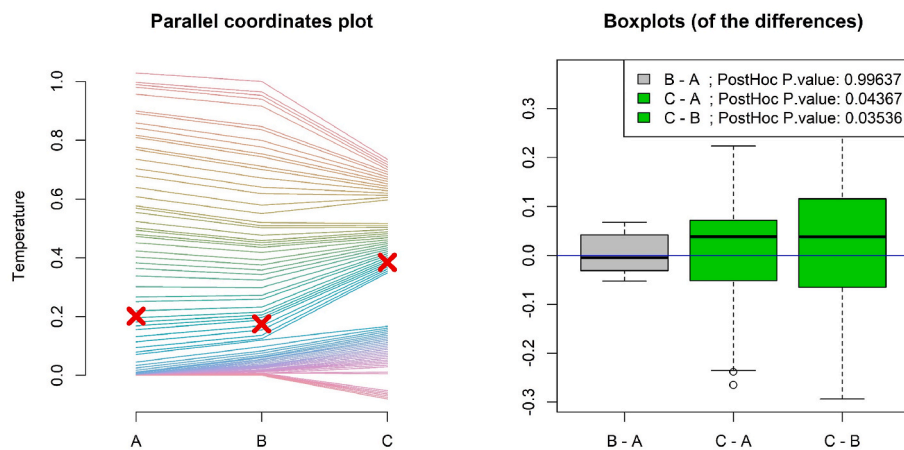


Fig. 17. Results of Friedman's test followed by post-hoc Nemenyi test comparing the temperature solutions computed via the analytical method (Hosseini et al., 2016) (A), gPINN with $w_{grad(k)_{x,t}} = 0.001$ (B), and gPINN with $w_{grad(k)_{x,t}} = 0.1$ (C).

distributions for three groups A, B, and C. It is worth noting that A, B, and C signify the same methods considered for Fig. 15. According to the p-values and boxplots, the results emphasize that there is no significant difference between the solutions calculated via the analytical method (Sherief and Saleh, 2005) and gPINN with $w_{grad(k)_{x,t}} = 0.001$. These

findings are reinforced by the parallel coordinates plot shown on the left-hand side of Fig. 16.

Finally, Fig. 17 concurs with our earlier findings. As proposed by the p-values, boxplots, and parallel coordinates plot, the solutions achieved by the analytical method (Hosseini et al., 2016) (A) and gPINN with

$w_{grad(k)_{x,t}} = 0.001$ follow the same distribution, and no significant difference is identified via the post-hoc Nemenyi test.

4.2. Second example

The second example takes the same geometry as the first and assumes that the boundary conditions are nonhomogeneous and that the elastic strip is subjected to both time-dependent mass concentration and temperature shock loading:

$$U(X=0, t) = 0, U(X=1.5, t) = 0, \quad (55)$$

$$C(X=0, t) = c_1 H(t), C(X=1.5, t) = 0,$$

$$\theta(X=0, t) = \theta_1 H(t), \theta(X=1.5, t) = 0.$$

This example uses the same neural network architecture as the one presented previously. That is, a separated structure is taken into consideration, and values of other influential hyperparameters are obtained via the sensitivity analysis, which is partly listed in Table 2. Table 2 presents eight combinations of the hyperparameters. Boxplots of the sensitivity analysis are detailed in Figs. 18–21. According to Table 2 and Figs. 18–21, the order of loss in scenario E is $O(10^{-4})$, which consequently provides the optimum set of hyperparameters for the given problem. Moreover, the weights assigned to PDE gradients are $w_{grad(k)_{x,t}} = 0.001$. In other words, a separate gPINN architecture with four hidden layers, each with 40 neurons and the learning rate of $1e-3$ achieves the lowest loss after 40,000 epochs.

In the following, trial solutions are formulated:

$$\widehat{C}(x, t; \Phi) = \|t\|^2 \|1.5 - X\| \|X\| N_C(x, t, \Phi) + \frac{2}{3} \|1.5 - X\| H(t), \quad (56)$$

$$\widehat{U}(x, t; \Phi) = \|t\|^2 \|X\| \|1.5 - X\| N_u(x, t, \Phi),$$

$$\widehat{\theta}(x, t; \Phi) = \|t\|^2 \|1.5 - X\| \|X\| N_T(x, t, \Phi) + \frac{2}{3} \|1.5 - X\| H(t).$$

Evidently, a complete set of initial and boundary conditions are imposed via hard constraints. Variations of output variables (i.e., mass concentration, displacement, and temperature) are plotted in Figs. 22–24 for different points in the time domain. Firstly, Fig. 22 illustrates the variations of non-dimensional mass concentration along the dimensionless distance. Indicating the propagation of wave fronts of mass concentration, Fig. 22 proves that the obtained values are consistent with Talbot results presented in the literature (Hosseini et al., 2016). It is worth noting that an increase in time results in the diffusion distance.

Fig. 23 highlights the distribution of non-dimensional displacement along the x direction. As anticipated, boundary conditions are fully met, proving gPINN's high capabilities in recognizing the behavior of field variables in coupled PDEs.

Fig. 24 demonstrates the location of temperature wave fronts for different time points. Similar to previous results associated with mass concentration and displacement, boundary conditions on either side are well-matched. The overall behavior of the non-dimensional temperature

supports earlier findings (Sherief and Saleh, 2005), (Hosseini et al., 2016). Taken as a whole, Figs. 22–24 reveal that field variables are reasonably smooth throughout the longitude axis, and no noticeable fluctuations are found. Similar to the results of the first example, wave fronts of all these field variables propagate at finite velocity along the x direction.

4.3. Third example

Third example of this section is inspired by the boundary conditions put forward by (Hosseini et al., 2016) with alteration of c_1 from 10 to 1.

$$\sigma(X=0, t) = 0, U(X=1.5, t) = 0, \quad (57)$$

$$C(X=0, t) = c_1 H(t), C(X=1.5, t) = 0,$$

$$\theta(X=0, t) = \theta_1 H(t), \theta(X=1.5, t) = 0.$$

According to the BCs mentioned above, the inner boundary of the strip is subjected to shock loading for both mass concentration and temperature. This loading is defined via the Heaviside unit step function of time. Identical to what has been suggested previously as enforcing boundary conditions, to satisfy some of BCs and all the ICs and facilitate the training procedure for gPINN, hard constraints are employed:

$$\widehat{C}(x, t; \Phi) = \|t\|^2 \|1.5 - X\| \|X\| N_C(x, t, \Phi) + \frac{2}{3} \|1.5 - X\| H(t), \quad (58)$$

$$\widehat{U}(x, t; \Phi) = \|t\|^2 \|1.5 - X\| \|X\| N_u(x, t, \Phi), \quad (59)$$

$$\widehat{\theta}(x, t; \Phi) = \|t\|^2 \|1.5 - X\| \|X\| N_T(x, t, \Phi) + \frac{2}{3} \|1.5 - X\| H(t). \quad (60)$$

Similar to previous examples, the weights of PDE gradients are $w_{grad(k)_{x,t}} = 0.001$. The last systematic sensitivity analysis to reach acceptable values for hyperparameters is partially indicated in Table 3 and Figs. 25–28. Table 3 outlines nine combinations of the hyperparameters. On account of the accurate results obtained via separate architectures in previous scenarios, investigations into the performance of a single neural network in this example can be neglected. Therefore, Table 3 concentrates on separate structures with different hyperparameters.

It is apparent that scenario "T" leads to the minimum loss value. As a result, three separate neural networks, all with the learning rate of $1e-3$, consist of five hidden layers and 60 neurons construct the gPINN structure. It seems that after 50,000 epochs, satisfactory results are achieved. This optimum setting is identical to the first example previously discussed in Table 1 (i.e., scenario L).

Variations of the non-dimensional mass concentration, displacement, and temperature within the specified limits are depicted in Figs. 29–31.

Fig. 29 presents the propagation of dimensionless concentration within X at a finite speed. At the same time, boundary conditions are fully met, which stresses the gPINN as a potent tool. Fig. 30 shows the distribution of dimensionless displacement along the X distance. It pinpoints the fact that incrementing the time will cause the diffusion

Table 2
Systematic sensitivity analysis of gPINN hyperparameters.

Scenario	#Layers	#Neurons	Learning rate	#Epochs	PINN architecture	Loss
A	2	20	1e-3	30,000	Separated	0.045410384
B	3	20	1e-3	30,000	Separated	0.02487797
C	3	20	1e-4	30,000	Separated	0.023485809
D	4	30	1e-3	30,000	Separated	0.004513041
E	4	40	1e-3	40,000	Separated	0.000299856
F	4	40	1e-4	40,000	Separated	0.016950223
G	4	50	1e-4	40,000	Separated	0.005336205
H	5	60	1e-3	50,000	Separated	0.002745898

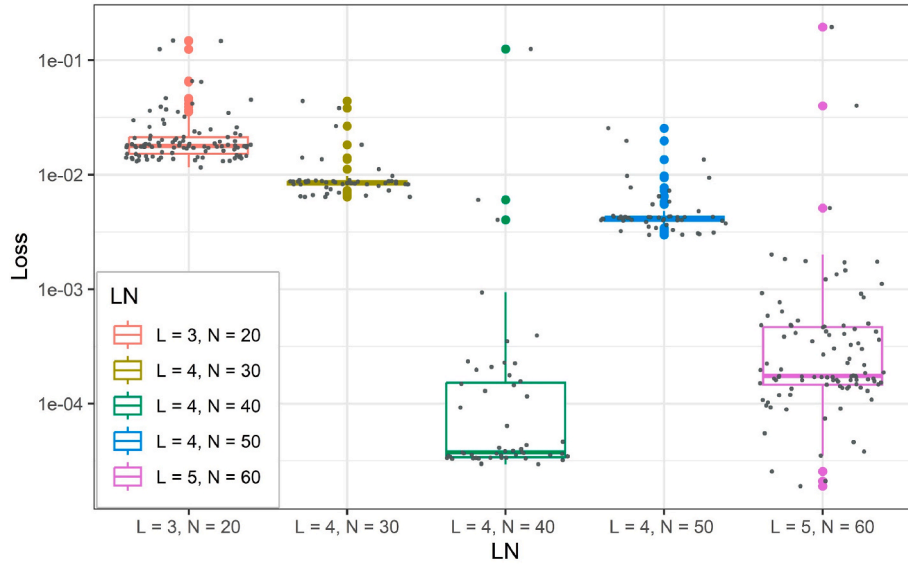


Fig. 18. Boxplot of sensitivity analysis exploring loss variations with respect to the number of layers and number of neurons.

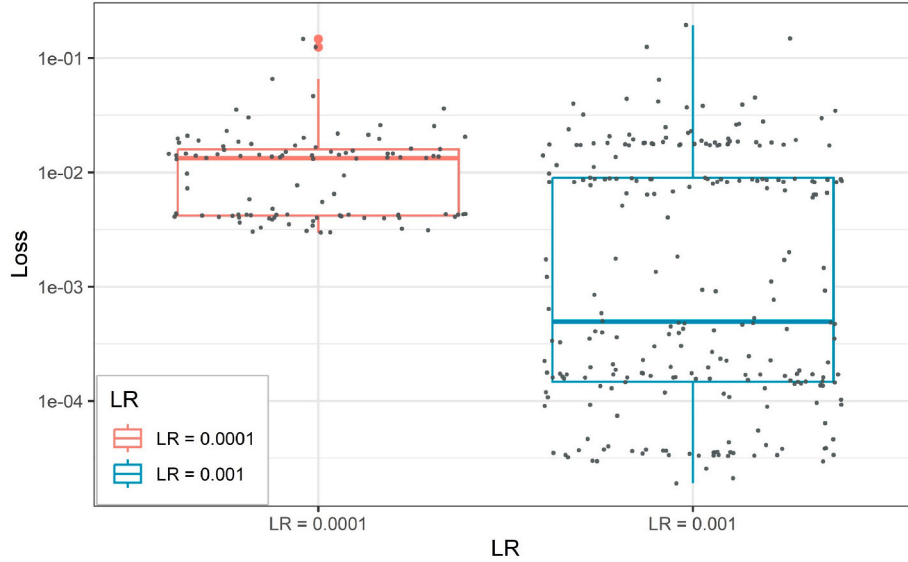


Fig. 19. Boxplot of sensitivity analysis studying the impact of learning rate on loss values.

distance to increase steadily. Fig. 31 addresses the distribution of dimensionless temperature. Similarly, propagation with finite velocity and satisfaction of boundary conditions are verified.

As illustrated in Fig. 32, values of the stress field are plotted within X distance. It can be seen that the stress propagation has a smooth pattern, and no unnormal disturbances are detected.

4.4. Fourth example

This example examines the application of gPINN to a two-dimensional generalized thermoelastic diffusion problem. A two-dimensional homogeneous and isotropic body is assumed to undergo both thermal and chemical potential shocks, and the governing equations are introduced by (Sherief et al., 2004), (Sherief and Saleh, 2005). For reasons of space, the formulations are not presented again since the functions are practically the same as previously presented examples except that they depend on X and Y . It is worthwhile mentioning that displacement is composed of two components:

$$U_X = U(X, Y, t), U_Y = U(X, Y, t), U_Z = 0. \quad (61)$$

The cubical dilatation $e = e_{kk}$ is also reformulated below:

$$e = \frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y}. \quad (62)$$

This example considers an interval of $[0, 3]$ for both of spatial variables (i.e., X and Y). Fig. 33 demonstrates the region in which the system of coupled PDEs is defined. OA and OC are equal to 3.0 and measure the non-dimensional distance of the x-axis and y-axis, respectively.

Initial conditions are homogenous and they are expressed as follows:

$$U(X, t=0) = V(X, t=0) = \theta(X, t=0) = P(X, t=0) = 0, \quad (63)$$

$$\dot{U}(X, t=0) = \dot{V}(X, t=0) = \dot{\theta}(X, t=0) = \dot{P}(X, t=0) = 0.$$

In addition, the thermal shock and the chemical shock loadings are in the following form:

$$\theta = \theta_0 H(t) H(L - |y|), \quad (64)$$

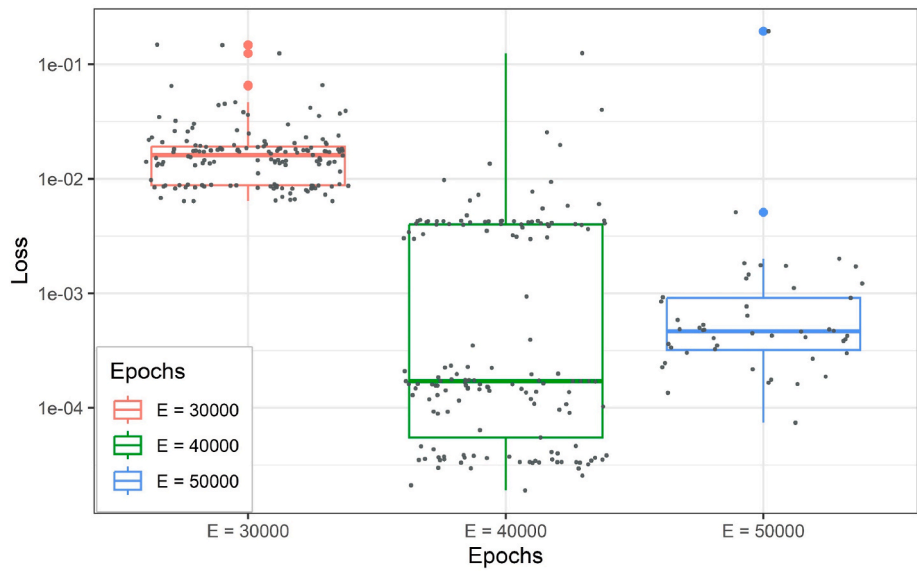


Fig. 20. Box plot of sensitivity analysis highlighting the influence of the number of epochs on loss values.

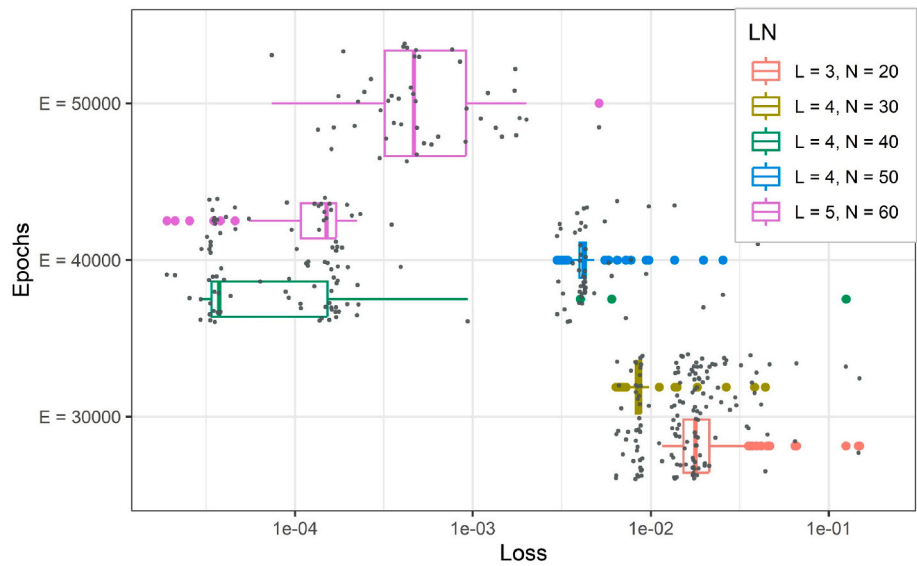


Fig. 21. Box plot of sensitivity analysis focusing on simultaneous impact of number of epochs and hidden layers/neurons on loss values.

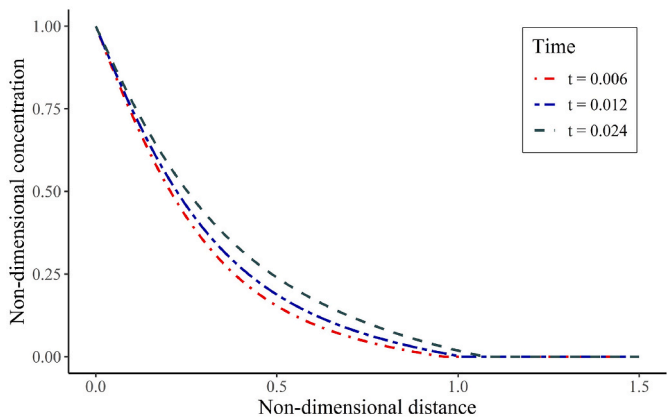


Fig. 22. Variations of non-dimensional mass concentration along X distance.

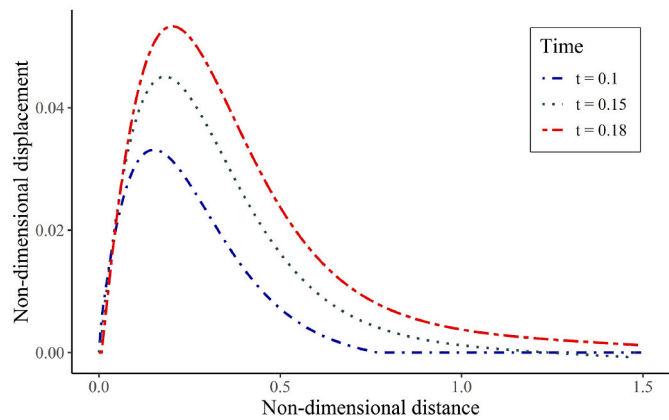


Fig. 23. Variations of non-dimensional displacement along X distance.

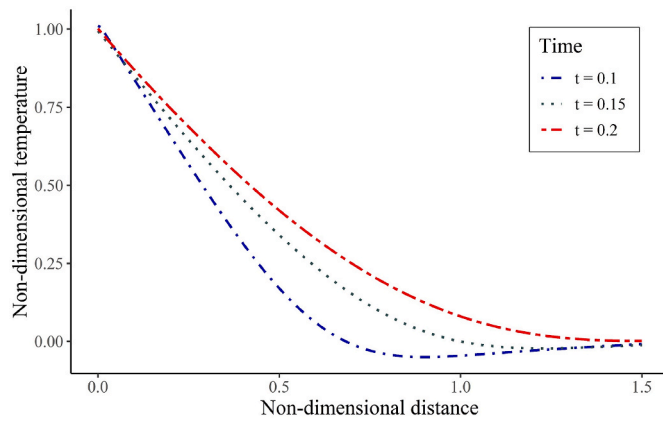


Fig. 24. Variations of non-dimensional temperature along X distance.

$$P = P_0 H(t) H(L - |y|),$$

where $L = OD = 0.2$, θ_0 and P_0 are constants and equal to 1.0.

The choice of neural networks' hyperparameters is an integral part of achieving satisfactory results. Different sets of neural network hyperparameters, including number of layers, number of neurons, and number of epochs are specified in Table 4. In light of the highly accurate results observed in previous examples, in all cases the tanh is selected as the activation function, the learning rate is equal to 10^{-3} , and every output variable (i.e., C, U, V , and θ) has its own neural network. Another key point to emphasize is that the gradients of PDE residuals adopt 0.001

as their weights. In order to assess the gPINN's accuracy in each scenario, the last four columns in Table 4 compute the L^2 relative error of the nondimensional output variables P, U, V , and θ , which are the chemical potential, horizontal displacement along OA, vertical displacement along OC, and temperature, respectively. L^2 relative error provides a credible criterion to measure the accuracy of predicted solutions and it is evaluated as follows (Nguyen-Thanh et al., 2021), (Wu et al., 2023):

$$L^2 \text{ relative error of } u = \frac{\|\hat{u} - u\|_2}{\|u\|_2} \quad (65)$$

Here, $\hat{u}$ signifies the solution obtained via gPINN and u is the actual value computed by analytical or conventional numerical methods. The gPINN results are compared with the hybrid Laplace transform finite element method (He et al., 2015a, 2015b).

Table 4 implies that an architecture with four hidden layers and each with 40 neurons can give reasonable accuracy after 80,000 iterations. In other words, scenario (C) achieves the lowest magnitude of L^2 relative error except for the temperature which is slightly higher than the scenario (D). Graphical comparisons of gPINN solutions with those reported in (He et al., 2015a, 2015b) are presented in Figs. 34–37.

Fig. 34 illustrates the variations of non-dimensional chemical potential along OA (i.e., x-axis) at dimensionless time $t = 0.1$. Figs. 35 and 36 display the distributions of non-dimensional horizontal and vertical components of displacement, respectively. The behavior of non-dimensional temperature along OA (i.e., $y = 0$) at dimensionless time $t = 0.06$ can also be seen in Fig. 37. The results of the two methods (i.e., gPINN and hybrid Laplace transform finite element method) in Figs. 34–37 demonstrate similar accuracy and indicate that gPINN is a

Table 3
Systematic sensitivity analysis of several sets of gPINN hyperparameters.

Scenario	#Layers	#Neurons	Learning rate	#Epochs	PINN architecture	Loss
A	2	20	1e-3	30,000	Separate	0.0129736
B	3	20	1e-3	30,000	Separate	0.01602337
C	3	20	1e-4	30,000	Separate	0.157284098
D	4	30	1e-3	30,000	Separate	0.044283565
E	4	40	1e-3	40,000	Separate	0.02247136
F	4	40	1e-4	40,000	Separate	0.013085827
G	4	50	1e-4	40,000	Separate	0.04483729
H	5	60	1e-3	40,000	Separate	0.004774155
I	5	60	1e-3	50,000	Separate	0.000685013

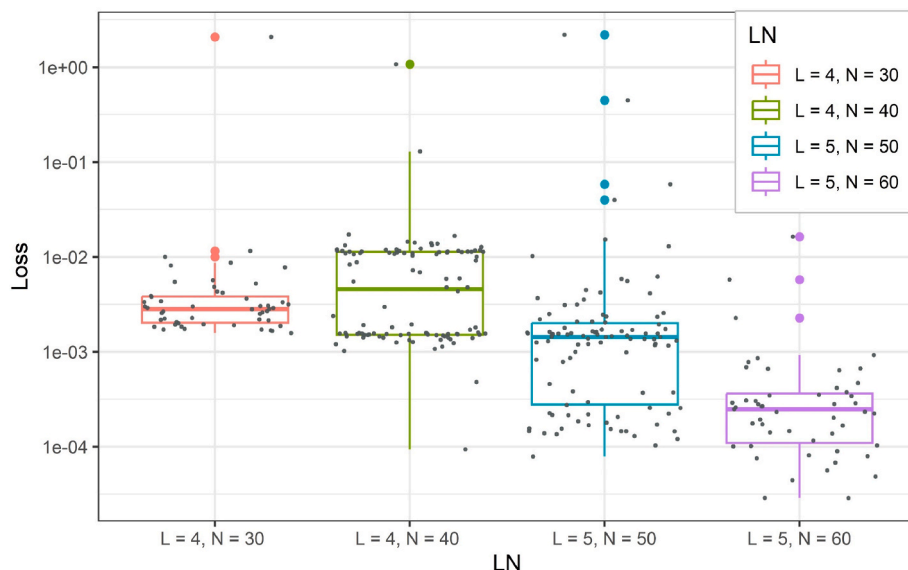


Fig. 25. Boxplot of sensitivity analysis revealing the impact of the number of hidden layers and neurons on loss values.

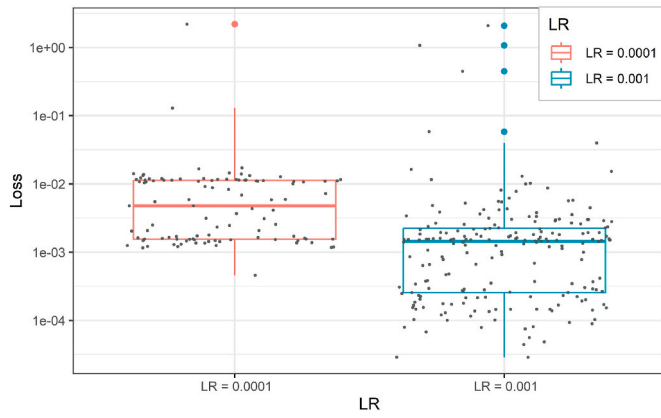


Fig. 26. Boxplot of sensitivity analysis comparing losses for two values of learning rate.

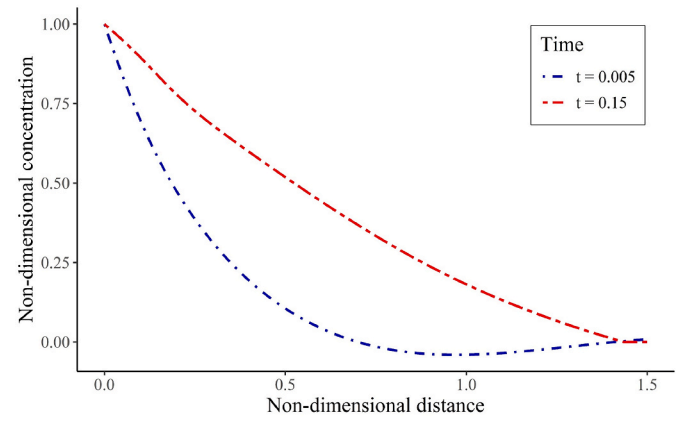


Fig. 29. Variations of non-dimensional concentration along X distance at different time instants.

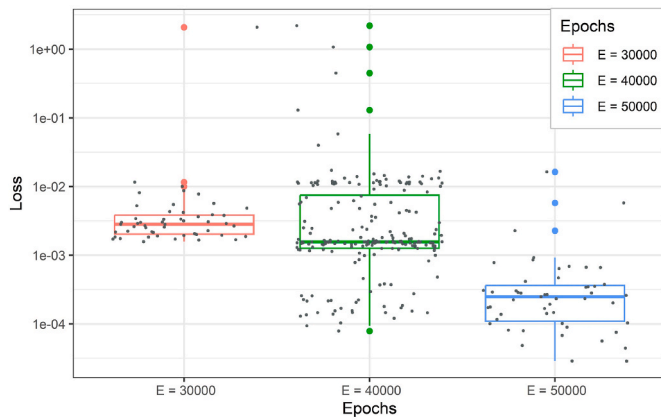


Fig. 27. Box plot of sensitivity analysis summarizing the effect of the number of epochs on total loss function.

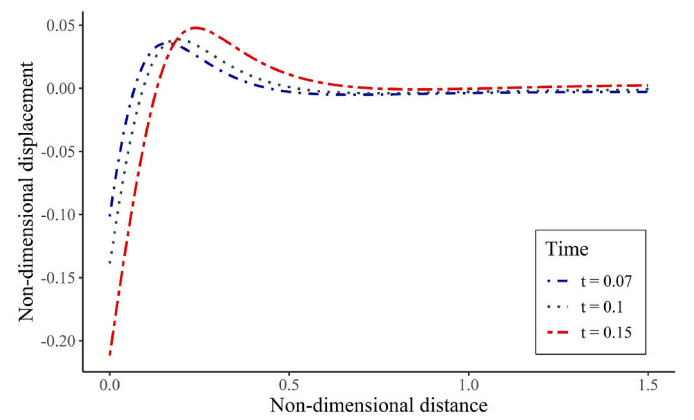


Fig. 30. Variations of non-dimensional displacement along X distance at different time instants.

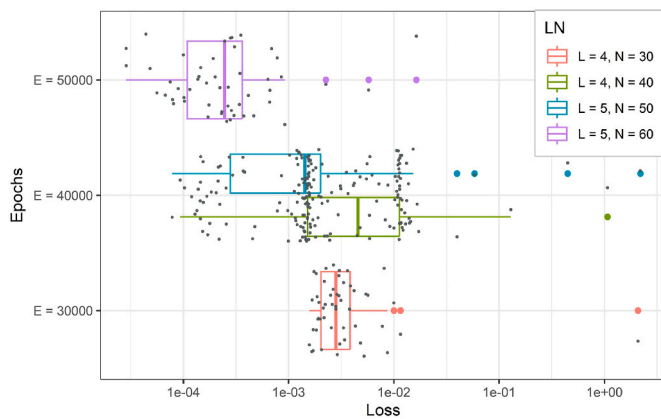


Fig. 28. Box plot of sensitivity analysis indicating variations of losses with respect to the number of epochs, hidden layers, and neurons.

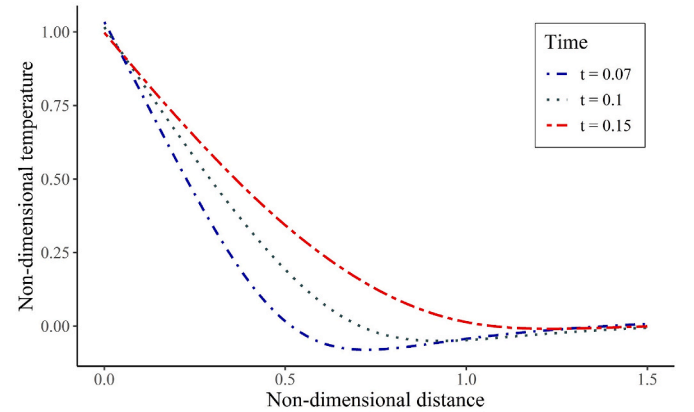


Fig. 31. Variations of non-dimensional temperature along X distance at different time instants.

viable option for solving such problems in higher dimensions. Furthermore, gPINN offers the potential for automation, allowing the model to be re-used for different combinations of parameters. Additionally, it can be used to explore the behavior of complex systems at various time points with a wide range of parameters.

5. Comparison of the computational costs

This research clearly has some limitations. The most critical one lies in the fact that extra computations are carried out to obtain the derivatives of loss terms with respect to the independent variables. Yu et al. (2022) proposed a metric referred to as the “relative computational cost” of gPINN to PINN. This criterion measures the additional training time imposed by gPINN. The following formula indicates how to

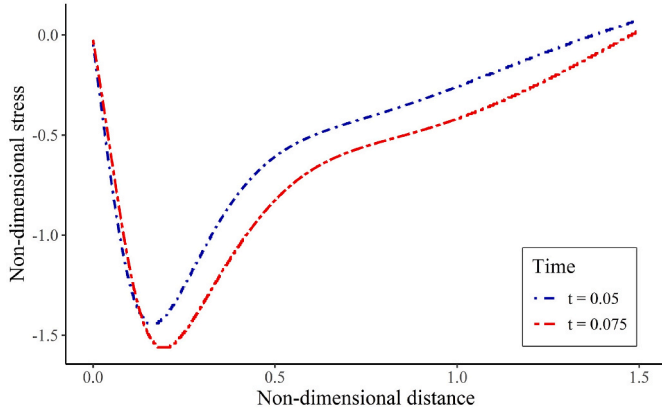


Fig. 32. Variations of non-dimensional stress along X distance at different time instants.

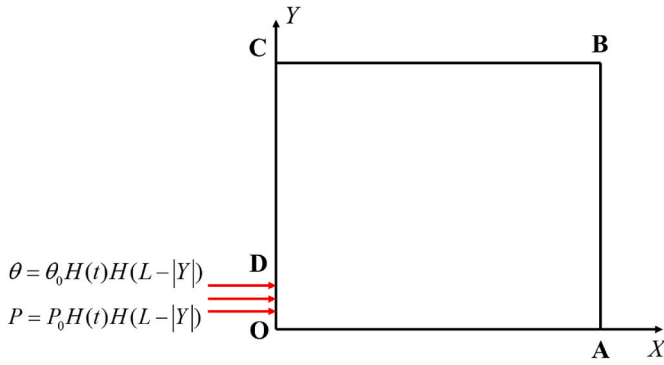


Fig. 33. Schematic of the two-dimensional half-space presented in the fourth example.

calculate this metric:

$$\text{relative computational cost} = \frac{\text{training time of the gPINN}}{\text{training time of the PINN}}. \quad (61)$$

Regarding the same number of training points for both structures of the gPINN and PINN, it is obvious that this proportion is greater than one, and its value highly depends on the dimensions and the complexity of the problem (Yu et al., 2022). In other words, as the dimension of the problem increases, more terms composed of gradients of loss terms with respect to inputs should be integrated into the gPINN.

In order to determine additional calculations imposed by gPINN, Table 5 details the values of the relative computational cost for every single example presented in this paper. It should be noted that the same number of training points is considered for PINN and gPINN to ensure fair comparisons.

According to Table 5, the relative computational cost varies from 1.96 to 2.7. Despite the limitations of the computational costs, our findings do suggest that the presence of the gradients of PDEs significantly improves the accuracy of final solutions.

Table 4

Comparison of L^2 relative error for different neural network architecture.

Scenario	#Layers	#Neurons	#Epochs	L^2 relative error			
				P	U	V	θ
A	3	30	40,000	≈7%	≈4%	≈5%	≈4%
B	3	40	50,000	≈2%	≈3%	≈3%	≈1%
C	4	40	80,000	≈0.1%	≈0.14%	≈0.2%	≈0.3%
D	4	50	100,000	≈0.5%	≈1%	≈2%	≈0.25%

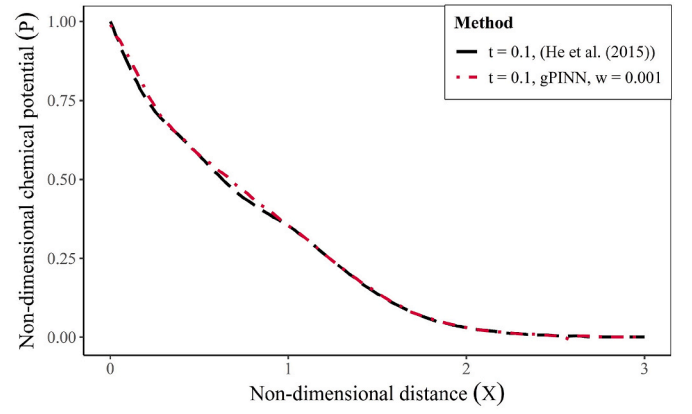


Fig. 34. Variations of non-dimensional chemical potential along x for y = 0 at t = 0.1.

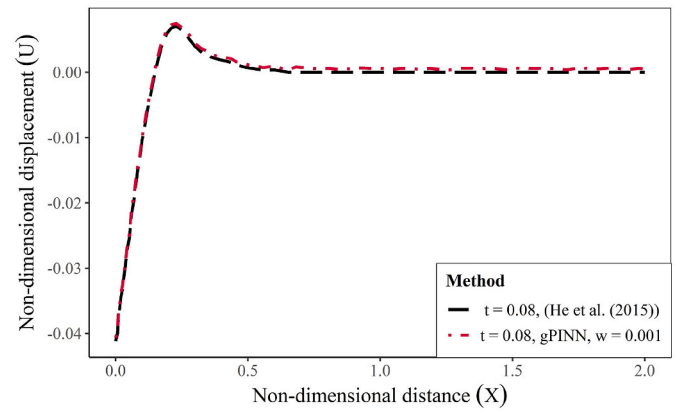


Fig. 35. Behaviour of non-dimensional horizontal displacement along x for y = 0 at t = 0.08.

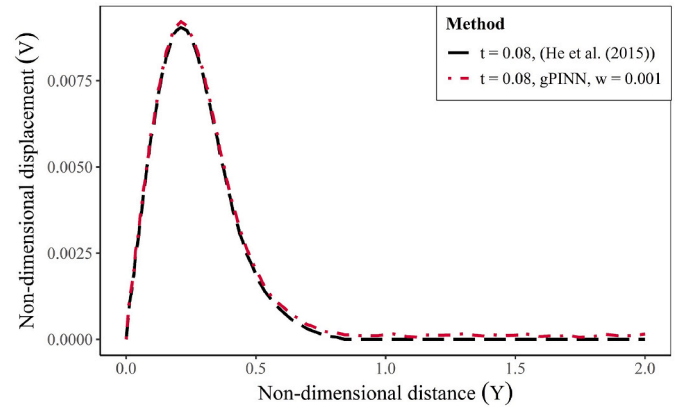


Fig. 36. Behaviour of non-dimensional vertical displacement along y for x = 0 at t = 0.08.

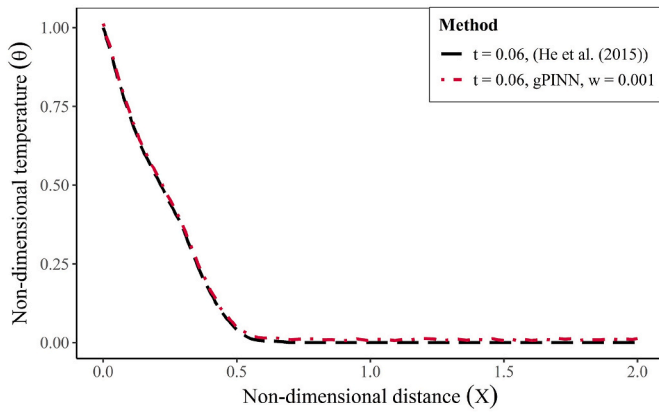


Fig. 37. Variations of non-dimensional temperature along x for $y = 0$ at $t = 0.06$.

Table 5

Details on the relative computational cost of gPINN to PINN for various examples of the generalized non-Fickian/non-Fourierian diffusion thermoelasticity.

Problem	Relative computational cost
First example	1.96
Second example	2.1
Third example	2.45
Fourth example	2.7

6. Conclusion

This paper seeks to extend the concept of a newly developed deep learning approach called gradient-enhanced physics-informed neural network (gPINN) to implement it on a system of coupled partial differential equations (PDEs). This research develops an innovative solution method for the generalized non-Fickian/non-Fourierian diffusion thermoelasticity. The problem is defined on a strip constructed of copper, and shock loadings are imposed in the form of the time-dependent Heaviside unit step function. A significant difference between gPINN with the original PINN is that it also includes gradients of the PDE with respect to input variables.

We believe that a customized version of this methodology is a valuable alternative to many conventional methods that are rife with inefficiencies. Of particular interest is that gPINN output and neural network parameters can be saved and later be utilized as a “surrogate model” for similar problems with minor discrepancies. This will assist the neural network in accelerating the training process. Other positive aspects of the gPINN include the following:

- One of the main shortcomings of commonly used numerical methods is that they require labor-intensive mesh generation calculations. Meanwhile, gPINN offers a mesh-free collocation method that enables faster and more efficient computations and modeling of various domains.
- An extrapolation capability is another powerful feature of gPINN that allows accurate predictions beyond the given domain.
- Choosing the appropriate weights for BC and IC losses can sometimes be complex and time-consuming. In gPINN and other PINN-based methods, the solutions can be modified so that BCs and ICs are precisely satisfied.
- In contrast to supervised learning, where access to large datasets is mandatory, gPINN performs optimally by simply taking into account the governing equations, BCs, and ICs.

The present findings have gone some way towards enhancing our

understanding of gPINN capacities. A brief review of findings is listed below:

- Due to the different behavior of output variables (i.e., concentration, displacement, and temperature), three separate neural networks are embodied in gPINN.
- To systematically tackle the issue of boundary and initial conditions satisfaction, some problematic ones are enforced as hard constraints.
- Obtained results further strengthened our confidence in the potential of gPINN. The most remarkable observation to emerge from the results is that gPINN completely learned the transient behavior of field variables. This is verified via three different scenarios of boundary conditions.
- Considerable insight has been gained with regard to wave propagation of field variables in finite velocity.
- The effect of relaxation time on the behavior of non-dimensional concentration is studied in the first example. It is worth noting that there is a satisfactory agreement between gPINN results and available observations obtained in previous studies.
- Comprehensive sensitivity analyses are implemented to rationalize the decision behind hyperparameters' values.
- Weights associated with loss of PDE gradients might play a critical role in convergence and accuracy of the final results. This assumption is investigated for several values of weights.
- A nonparametric statistical test (Friedman's test followed by a posthoc Nemenyi test) confirms that the distributions obtained by gPINN with $w_{grad(k)_{x,t}} = 0.001$ and analytical solutions are identical.
- Application of gPINN to a two-dimensional thermoelastic diffusion problem revealed that the proposed method can predict the behavior of field variables in higher dimensions with reasonable accuracy. The gPINN model was able to capture the exact field variables with L^2 relative error of less than 0.3%. The results demonstrate the potential of gPINN to be used as an efficient and accurate tool for predicting complex physical phenomena.

Our investigations so far have only been applied to non-Fickian/non-Fourierian diffusion thermoelasticity. The results point to the likelihood that gPINN seems to perform well in similar systems of coupled PDEs. However, extreme caution must be exercised in high-dimensional problems because it requires the computation of gradients with respect to every single input variable. Consequently, this will be significantly time-consuming.

Another weak point that needs to be considered is that the training procedure in gPINN and in PINN-based methods is a non-convex multi-objective problem. Thus, the high chance of getting trapped in local optima is of concern. Neural networks' inability to find unique solutions is another issue that sometimes poses difficulty. In order to limit this uncertainty, multiple executions of the training procedure can be conducted and the solutions corresponding to the lowest loss can be selected.

A critical question to respond to for future research is the incorporation of stochastic parameters in governing equations. On a broader level, future research will concentrate on two issues:

- Optimal values of weights for losses of PDE gradients are obtained through sensitivity analysis. In cases where values of these weights are crucial, developing an algorithm that ensures optimality might be helpful.
- Various hyperparameters are defined for a neural network whose exact number varies based on the type of neural network. Values of some hyperparameters have a tremendous effect on accuracy and convergence rate. Adjusting these values is achieved experimentally. Developing an automatic and intelligent mechanism for detecting appropriate values is deferred to future work.

CRediT authorship contribution statement

Katayoun Eshkofti: Methodology, Software, Visualization, Investigation, Validation, Writing – original draft. **Seyed Mahmoud Hosseini:** Supervision, Conceptualization, Methodology, Visualization, Investigation, Writing – review & editing. **Writing- Reviewing and Editing.**

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Abbreviations

2 TT	2-Temperature Theory of Thermoelasticity
AD	Automatic Differentiation
ANN	Artificial Neural Network
BC	Boundary Condition
BPINN	Bayesian Physics-Informed Neural Network
cPINN	Conservative Physics-Informed Neural Network
DEM	Deep Energy Method
DL	Deep Learning
FDM	Finite Difference Method
FEM	Finite Element Method
GFD	Generalized Finite Difference
GN	Green-Naghdi
gPINN	Gradient-Enhanced Physics-Informed Neural Network
IC	Initial Condition
MHD	Magnetohydrodynamics
MLPG	Meshless Local Petrov-Galerkin
NRPINN	New Reptile Initialization based PINN
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
P-DEM	Parametric Deep Energy Method
PI-GAN	Physics-Informed Generative Adversarial Network
PINN	Physics-Informed Neural Network
PI-PP-NN	Physics-Informed and Physics-Penalised Neural Network Model
XPINN	Extended Physics-Informed Neural Network

Nomenclature

$\mathcal{F}$	differential operator
Ω	domain of the problem
$\partial\Omega$	domain of the boundary condition
f	forcing term
$\mathbf{u}$	desired output vector
$\hat{\mathbf{u}}$	NN-based approximation of $\mathbf{u}$
ϑ	parameters of the problem
β	boundary conditions
$\mathbf{z}$	vector of independent spatio-temporal variables
$\mathcal{L}$	loss function (mean squared error)
σ_{ij}	stress tensor
e_{ij}	strain tensor
c	molar concentration
μ	chemical potential
C_{ijkl}	elastic coefficient
α_{ij}	mechanical diffusion constant
β	chemical potential constant
ρ	density
u	displacement
$\mu^{(a)}$	inertial chemical potential
I	diffusion source
c_0	reference concentration
J_i	diffusion flux
D_0	diffusion coefficient
γ	constant

T	absolute temperature
τ_0	relaxation time
λ, μ	lamé's constants
t	time
$N(x, t; \Phi)$	neural network with x and t as inputs and Φ as hyperparameters
θ	temperature
a	measure of thermodiffusion effect
b	measure of diffusive effect
α_t	coefficient of linear thermal expansion
α_c	coefficient of linear diffusion expansion
τ	diffusion relaxation time
k	thermal conductivity
c_E	specific heat
T_0	reference temperature
P_1, θ_1, C_1	constants for chemical potential, temperature, and concentration
$w_{grad(k)}_{x,t}$	weights assigned to gradients of k th PDE with respect to x and t

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